

Lecture7a

October 27, 2025

1 CS5489 - Machine Learning

2 Lecture 7a - Deep Learning

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3 Outline

- Going deeper
 - ReLU and Batchnorm
- Optimization methods
- Deep architectures and Image classification
- Transfer learning

```
[1]: # setup
%matplotlib inline
import matplotlib_inline # setup output image format
matplotlib_inline.backend_inline.set_matplotlib_formats('retina')
import matplotlib.pyplot as plt
plt.rcParams['figure.dpi'] = 100 # display larger images
import matplotlib
from numpy import *
import numpy as np
from sklearn import *
from scipy import stats
from scipy import signal
import random
import math
from prettytable import PrettyTable

rbow = plt.get_cmap('rainbow')
```

```
[2]: import torch
import torch.nn as nn
import torch.optim as optim
from torch.utils.data import DataLoader, TensorDataset
import struct
import sys
```

```

print(f"Python: {sys.version} PyTorch: {torch.__version__}")

# Set device
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
print(f"Using device: {device}")

```

Python: 3.13.5 | packaged by Anaconda, Inc. | (main, Jun 12 2025, 16:09:02) [GCC 11.2.0] PyTorch: 2.8.0+cu129
Using device: cuda

```

[3]: def plot_history(history):
    fig, ax1 = plt.subplots()

    ax1.plot(history['train_loss'], 'r', label="training loss ({:.6f})".
    ↪format(history['train_loss'][-1]))
    ax1.plot(history['val_loss'], 'r--', label="validation loss ({:.6f})".
    ↪format(history['val_loss'][-1]))
    ax1.grid(True)
    ax1.set_xlabel('iteration')
    ax1.legend(loc="best", fontsize=9)
    ax1.set_ylabel('loss', color='r')
    ax1.tick_params('y', colors='r')

    if 'train_accuracy' in history:
        ax2 = ax1.twinx()

        ax2.plot(history['train_accuracy'], 'b', label="training acc ({:.4f})".
        ↪format(history['train_accuracy'][-1]))
        ax2.plot(history['val_accuracy'], 'b--', label="validation acc ({:.
        ↪4f})".format(history['val_accuracy'][-1]))

        ax2.legend(loc="best", fontsize=9)
        ax2.set_ylabel('acc', color='b')
        ax2.tick_params('y', colors='b')
    plt.show()

```

```

[4]: def show_imgs(W_list, nc=10, highlight_green=None, highlight_red=None,
    ↪titles=None):
    nfilter = len(W_list)
    nr = (nfilter - 1) // nc + 1
    for i in range(nr):
        for j in range(nc):
            idx = i * nc + j
            if idx == nfilter:
                break
            plt.subplot(nr, nc, idx + 1)
            cur_W = W_list[idx]

```

```

plt.imshow(cur_W, cmap='gray', interpolation='nearest')
if titles is not None:
    plt.title(titles % idx)

if ((highlight_green is not None) and highlight_green[idx]) or \
    ((highlight_red is not None) and highlight_red[idx]):
    ax = plt.gca()
    if highlight_green[idx]:
        mycol = '#00FF00'
    else:
        mycol = 'r'
    for S in ['bottom', 'top', 'right', 'left']:
        ax.spines[S].set_color(mycol)
        ax.spines[S].set_lw(2.0)
    ax.xaxis.set_ticks_position('none')
    ax.yaxis.set_ticks_position('none')
    ax.set_xticks([])
    ax.set_yticks([])
else:
    plt.gca().set_axis_off()

```

```

[5]: def read_32int(f):
    return struct.unpack('>I', f.read(4))[0]
def read_img(img_path):
    with open(img_path, 'rb') as f:
        magic_num = read_32int(f)
        num_image = read_32int(f)
        n_row = read_32int(f)
        n_col = read_32int(f)
        #print 'num_image = {}; n_row = {}; n_col = {}'.format(num_image,
↪n_row, n_col)
        res = []
        npixel = n_row * n_col
        res_arr = fromfile(f, dtype='B')
        res_arr = res_arr.reshape((num_image, n_row, n_col), order='C')
        #print 'image data shape = {}'.format(res_arr.shape)
        return num_image, n_row, n_col, res_arr
def read_label(label_path):
    with open(label_path, 'rb') as f:
        magic_num = read_32int(f)
        num_label = read_32int(f)
        #print 'num_label = {}'.format(num_label)
        res_arr = fromfile(f, dtype='B')
        #print res_arr.shape
        #res_arr = res_arr.reshape((num_label, 1))
        res_arr = res_arr.ravel()
        #print 'label data shape = {}'.format(res_arr.shape)

```

```
return num_label, res_arr
```

```
[6]: n_train, nrow, ncol, training = read_img('data/train-images.idx3-ubyte')
_, trainY = read_label('data/train-labels.idx1-ubyte')
n_test, _, _, testing = read_img('data/t10k-images.idx3-ubyte')
_, testY = read_label('data/t10k-labels.idx1-ubyte')

# for demonstration we only use 10% of the training data
sample_index = range(0, training.shape[0], 10)
training = training[sample_index]
trainY = trainY[sample_index]
print(training.shape)
print(trainY.shape)
print(testing.shape)
print(testY.shape)
```

```
(6000, 28, 28)
(6000,)
(10000, 28, 28)
(10000,)
```

```
[7]: # Convert to PyTorch tensors and scale to 0-1
trainI = torch.from_numpy(training.reshape((6000, 1, 28, 28)) / 255.0).float()
testI = torch.from_numpy(testing.reshape((10000, 1, 28, 28)) / 255.0).float()

print(trainI.shape)
print(testI.shape)
```

```
torch.Size([6000, 1, 28, 28])
torch.Size([10000, 1, 28, 28])
```

```
[8]: # Generate fixed validation set
from sklearn.model_selection import train_test_split

vtrainI_np, validI_np, vtrainY_np, validY_np = train_test_split(
    trainI.numpy(), trainY,
    train_size=0.9, test_size=0.1, random_state=4487
)

vtrainI = torch.from_numpy(vtrainI_np).float()
validI = torch.from_numpy(validI_np).float()
vtrainY = torch.from_numpy(vtrainY_np).long()
validY = torch.from_numpy(validY_np).long()

validsetI = (validI, validY)

print(f"Training set: {vtrainI.shape}, {vtrainY.shape}")
```



```
print(f"Validation set: {validI.shape}, {validY.shape}")

cid, cfreq = vtrainY.unique(return_counts=True)

print(f"Training set Distribution:", cfreq.numpy().tolist())
```

```
Training set: torch.Size([5400, 1, 28, 28]), torch.Size([5400])
Validation set: torch.Size([600, 1, 28, 28]), torch.Size([600])
Training set Distribution: [491, 599, 554, 556, 513, 518, 567, 584, 496, 522]
```

4 Problems with Going Deeper

- **Vanishing Gradient Problem 1**
 - successive multiplications of small gradients gives smaller gradients, and converges to 0
 - the gradients backpropagated to the first few layers has small signal.
- Example: for 4-layers,

$$\frac{dL}{da_j} = \frac{dg_1^T}{da_j} \frac{dz_1^T}{dg_1} \frac{dg_2^T}{dz_1} \frac{dz_2^T}{dg_2} \frac{dg_3^T}{dz_2} \frac{dz_3^T}{dg_3} \frac{dg_4^T}{dz_3} \frac{df^T}{dg_4} \frac{dL}{df}$$

- **Vanishing Gradient Problem 2**
 - using backprop, the gradient at a node is the summation over $O(H^D)$ paths
 - * D is the number of layers to the output layer.
 - * H is the number of nodes in the layer (assuming they all have the same width).
 - the original loss signal gets “washed out”.
- **Dataset Size**
 - a “small” network with just 40 inputs, 30 hidden nodes, and 1 output has ~1200 parameters.
 - if we don’t have enough samples:
 - * large variance in the parameter estimator (what you get may be far from the truth)
 - * deeper networks are more complex, which are easier to overfit the training data.
- *How many samples do we need?*
 - Theorem (Bartlett, Maierov, Meir, 1998) > Suppose N is a feed-forward network with W weights, L layers, and all non-output gates having a fixed piecewise-polynomial activation function with a fixed number of pieces (e.g., ReLU). Then $VCdim(N) = O(WL \log W + WL^2)$.
- VC dimension is a statistical measure of function class capacity:
 - to achieve certain level of generalization, sample size needs to scale proportionally to VC dim.
 - for the same number of parameters, the deeper network requires more data (WL^2).
 - increasing the number of weights, requires a super-linear increase in sample size ($W \log W$).

5 ReLU activation function

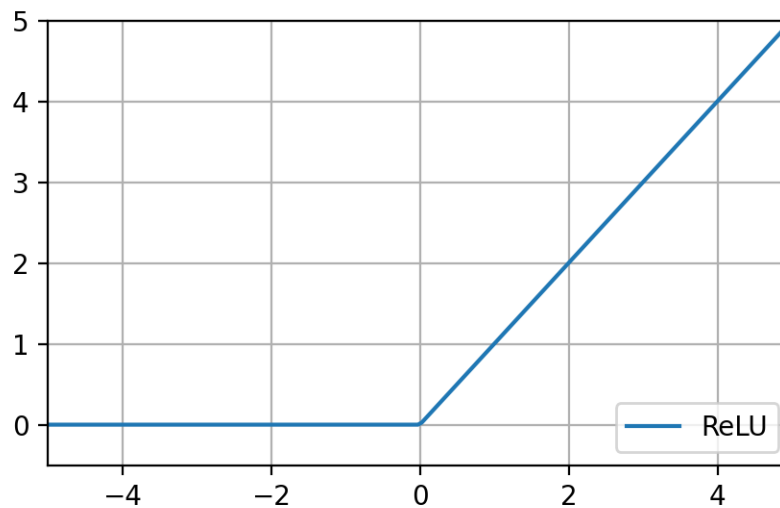
- Rectified Linear Unit: $\text{ReLU}(z) = \max(0, z)$
 - easier to train with: gradient is either 0 or 1.
 - faster: don't need to calculate exponential
 - sparse representation: most nodes will output zero.

```
[9]: x=linspace(-5,5,200)
fs = {'ReLU': lambda x: maximum(0,x),
      }

actfig = plt.figure(figsize=(5,3))
for name,f in fs.items():
    plt.plot(x, f(x), label=name)
plt.legend(loc="lower right")
plt.axis([-5,5,-0.5,5])
plt.grid(True)
plt.close()
```

```
[10]: actfig
```

```
[10]:
```



6 Advantage of Sparsity

- if a hidden node $h = 0$, then $\frac{dL}{dh} = 0$.
 - This blocks some paths when computing the gradients.
 - Gradient signal is less washed out.
 - Reduces the vanishing gradient problem.

7 Better Network Parameterization

- There are equivalent parameterizations of the network by scaling up/down the weights in successive layers.
 - $f(\mathbf{x}) = \mathbf{A}^T r(\mathbf{B}^T \mathbf{x}) = \frac{1}{\epsilon} \mathbf{A}^T r(\epsilon \mathbf{B}^T \mathbf{x})$
 - * $\epsilon > 0$ and $r(\cdot)$ is the ReLU activation.
- **Problem:**
 - “internal covariate shift” - change in the distribution of activations during training, due to changes in the parameters.

8 Why is it bad?

- suppose, we have a linear network:
- $y = xw_1w_2w_3w_4$
- gradient of each layer is $(g_1, \dots, g_4)$
- update the parameters with GD:
 - $y = x(w_1 - \eta g_1)(w_2 - \eta g_2)(w_3 - \eta g_3)(w_4 - \eta g_4)$
 - * there are many higher-order terms, e.g., $w_1w_2\eta^2g_3g_4$
 - although w_i are updated independently, they strongly affect each other.
 - hence, if the distribution of activations changes in 1 layer, then all layers are affected, and we need to adjust other layers.

9 Solution: Batch Normalization

- For each node in each layer, normalize the outputs to zero mean and unit variance, *over each mini-batch*.
 - this is analogous to the idea of normalizing the input feature vector to (0,1) Gaussian with standard ML models!
- Place batchnorm layer after linear transformation.
 - Let $\{y_i\}_{i=1}^N$ be the output of the linear transform in one minibatch.
 - For each node (dimension) in the layer:
 - * normalize: $\hat{y}_i = \frac{y_i - \mu}{\sqrt{\sigma^2 + \epsilon}}$
 - μ, σ^2 are the mean and variance of $\{y_i\}$ in the mini-batch.
 - ϵ is a small constant for numerical stability.
 - * scale-shift: $z_i = \gamma \hat{y}_i + \beta$
 - γ, β are learnable parameters
 - puts the output in the proper regime of the non-linear activation.
 - * The final distribution has mean β and variance γ^2 .
- Notes:
 - batchnorm is applied to each node independently.
 - should put the batchnorm layer after the linear transformation layer.
 - the bias of the linear layer is not necessary since it is removed by batchnorm
- Training:
 - gradients can be computed through the batchnorm layer as usual.
- Training effects:

- training is accelerated; can use higher learning rates
- more stable gradients during training
 - * increasing the scale of the activations decreases the gradient
 - * self-correcting stabilization.
- better generalization
 - * no need for dropout or L2 regularization.

10 Example: MNIST

- for each Conv2D/Dense layer:
 - change activation to linear (default); remove bias term
 - append batch-norm and ReLU activation

```
[11]: class DeepCNN(nn.Module):
    def __init__(self, weight_decay=0.0001):
        super(DeepCNN, self).__init__()
        self.weight_decay = weight_decay

        self.conv1 = nn.Conv2d(1, 10, kernel_size=5, stride=2, padding=2)
        self.conv2 = nn.Conv2d(10, 40, kernel_size=5, stride=2, padding=2)
        self.conv3 = nn.Conv2d(40, 80, kernel_size=5, stride=1, padding=2)
        self.fc1 = nn.Linear(80 * 7 * 7, 50) # Adjusted based on input size
        and strides
        self.fc2 = nn.Linear(50, 10)

        self.relu = nn.ReLU()
        self.dropout = nn.Dropout(0.5)
        self.softmax = nn.Softmax(dim=1)

    def forward(self, x):
        x = self.relu(self.conv1(x))
        x = self.relu(self.conv2(x))
        x = self.relu(self.conv3(x))
        x = x.view(x.size(0), -1) # Flatten
        x = self.relu(self.fc1(x))
        x = self.fc2(x)
        return x
```

```
[12]: # Early stopping implementation
class EarlyStopping:
    def __init__(self, patience=5, min_delta=0.0001, verbose=1):
        self.patience = patience
        self.min_delta = min_delta
        self.counter = 0
        self.best_accuracy = 0.0
        self.early_stop = False
        self.verbose = verbose
```

```

def __call__(self, current_accuracy):
    if current_accuracy - self.best_accuracy > self.min_delta:
        self.best_accuracy = current_accuracy
        self.counter = 0
    else:
        self.counter += 1
        if self.verbose:
            print(f'EarlyStopping counter: {self.counter} out of {self.
↪patience}')
        if self.counter >= self.patience:
            self.early_stop = True
    return self.early_stop

```

```

[13]: # Model Parameter Summarization
def summarize_layer_parameters(model):
    """
    Summarizes the number of parameters in each layer of a PyTorch model.
    """
    trainable_table = PrettyTable(["Module Name", "Parameters"])
    untrainable_table = PrettyTable(["Module Name", "Parameters"])
    total_params = {'train':0, 'untrain':0}
    for name, parameter in model.named_parameters():
        if parameter.requires_grad: # Only count trainable parameters
            params = parameter.numel()
            trainable_table.add_row([name, params])
            total_params['train'] += params
        else: # Only count untrainable parameters
            params = parameter.numel()
            untrainable_table.add_row([name, params])
            total_params['untrain'] += params

    print(trainable_table)
    print(f"\nTotal Trainable Parameters: {total_params['train']}")
    if total_params['untrain'] > 0:
        print(untrainable_table)
    print(f"\nTotal Un-Trainable Parameters: {total_params['untrain']}")

```

```

[14]: from torch.optim.lr_scheduler import ↪
↪ReduceLRonPlateau, CosineAnnealingLR, ExponentialLR, StepLR, MultiStepLR, LinearLR

# Model Evaluation
def evaluate_model(model, eval_loader, criterion=None):
    model.eval()
    eval_correct = 0
    eval_total = 0
    eval_loss = 0.0

```

```

with torch.no_grad():
    for data, target in eval_loader:
        data, target = data.to(device), target.to(device)
        output = model(data)
        _, predicted = torch.max(output.data, 1)
        eval_total += target.size(0)
        eval_correct += (predicted == target).sum().item()

        if criterion is not None:
            loss = criterion(output, target)
            eval_loss += loss.item()

eval_accuracy = 100 * eval_correct / eval_total
avg_eval_loss = eval_loss / len(eval_loader)

return eval_accuracy if criterion is None else (eval_accuracy,
↪avg_eval_loss)

# Model Training
def train_model(model, train_loader, valid_loader, criterion, optimizer,
↪epochs=100, patience=5, batch_scheduler=None, epoch_scheduler=None):
    history = {
        'train_loss': [], 'val_loss': [],
        'train_accuracy': [], 'val_accuracy': [],
        'learning_rate': []
    }
    early_stopping = EarlyStopping(patience=patience)
    assert batch_scheduler is None or epoch_scheduler is None

    # best_val_loss = float('inf')
    # patience_counter = 0

    for epoch in range(epochs):
        # Training phase
        model.train()
        train_loss = 0.0
        train_correct = 0
        train_total = 0

        for batch_idx, (data, target) in enumerate(train_loader):
            data, target = data.to(device), target.to(device)

            optimizer.zero_grad()
            output = model(data)

```

```

        # loss = criterion(output, torch.nn.functional.one_hot(target,
↪num_classes=10).float()) also works
        loss = criterion(output, target)

        loss.backward()
        optimizer.step()
        if batch_scheduler is not None:
            batch_scheduler.step()
            history['learning_rate'].append(optimizer.
↪state_dict()['param_groups'][0]['lr'])

        train_loss += loss.item()
        _, predicted = torch.max(output.data, 1)
        train_total += target.size(0)
        train_correct += (predicted == target).sum().item()

    train_accuracy = 100 * train_correct / train_total
    avg_train_loss = train_loss / len(train_loader)

    # Validation phase
    val_accuracy, avg_val_loss =
↪evaluate_model(model, valid_loader, criterion)

    # Store history
    history['train_loss'].append(avg_train_loss)
    history['val_loss'].append(avg_val_loss)
    history['train_accuracy'].append(train_accuracy / 100)
    history['val_accuracy'].append(val_accuracy / 100)

    train_msg = f' Train Loss: {avg_train_loss:.6f}, Train Acc:
↪{train_accuracy:.2f}%; '
    valid_msg = f' Val Loss: {avg_val_loss:.6f}, Val Acc: {val_accuracy:.
↪2f}%'

    print(f'Epoch {epoch+1}/{epochs}: ', train_msg, valid_msg)

    # Check early stopping
    if patience > 0:
        early_stopping(val_accuracy)
        if early_stopping.early_stop:
            print("Early stopping triggered")
            break
        else:
            torch.save(model.state_dict(), 'best_model.pth')
    else:
        torch.save(model.state_dict(), 'best_model.pth')

```

```

        if epoch_scheduler is not None:
            if isinstance(epoch_scheduler, ReduceLROnPlateau):
                epoch_scheduler.step(avg_val_loss)
            else:
                epoch_scheduler.step()
            history['learning_rate'].append(optimizer.
↪state_dict()['param_groups'][0]['lr'])

    # Load best model
    model.load_state_dict(torch.load('best_model.pth'))
    return history, model

```

```

[15]: # Initialize random seed
torch.manual_seed(4487)
random.seed(4487)
np.random.seed(4487)
if torch.cuda.is_available():
    torch.cuda.manual_seed(4487)
    torch.backends.cudnn.deterministic = True

# Create model instance
model = DeepCNN()

# Print model architecture
print(model)
summarize_layer_parameters(model)

# Move model to device (GPU if available)
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
model = model.to(device)
print(f"Using device: {device}")

# Prepare data - ensure correct shape for PyTorch (channel first)
# Assuming vtrainI, vtrainY, validI, validY are already defined as in previous
↪cells

# Convert from (batch, height, width, channels) to (batch, channels, height,
↪width)
if len(vtrainI.shape) == 4 and vtrainI.shape[-1] == 1: # If shape is (N, 28,
↪28, 1)
    vtrainI = vtrainI.transpose(0, 3, 1, 2)
    validI = validI.transpose(0, 3, 1, 2)

# Create DataLoaders
batch_size = 50
train_dataset = TensorDataset(vtrainI, vtrainY)

```



```

valid_dataset = TensorDataset(validI, validY)

train_loader = DataLoader(train_dataset, batch_size=batch_size, shuffle=True)
valid_loader = DataLoader(valid_dataset, batch_size=batch_size, shuffle=False)

# Compile the network (equivalent in PyTorch)
criterion = nn.CrossEntropyLoss() # This includes softmax + cross entropy
optimizer = optim.SGD(model.parameters(), lr=0.02, momentum=0.9, nesterov=True)

# Train the model
history, model = train_model(model, train_loader, valid_loader, criterion,
    ↪optimizer, epochs=100)

# Plot the training history
plot_history(history)

```

```

DeepCNN(
  (conv1): Conv2d(1, 10, kernel_size=(5, 5), stride=(2, 2), padding=(2, 2))
  (conv2): Conv2d(10, 40, kernel_size=(5, 5), stride=(2, 2), padding=(2, 2))
  (conv3): Conv2d(40, 80, kernel_size=(5, 5), stride=(1, 1), padding=(2, 2))
  (fc1): Linear(in_features=3920, out_features=50, bias=True)
  (fc2): Linear(in_features=50, out_features=10, bias=True)
  (relu): ReLU()
  (dropout): Dropout(p=0.5, inplace=False)
  (softmax): Softmax(dim=1)
)

```

Module Name	Parameters
conv1.weight	250
conv1.bias	10
conv2.weight	10000
conv2.bias	40
conv3.weight	80000
conv3.bias	80
fc1.weight	196000
fc1.bias	50
fc2.weight	500
fc2.bias	10

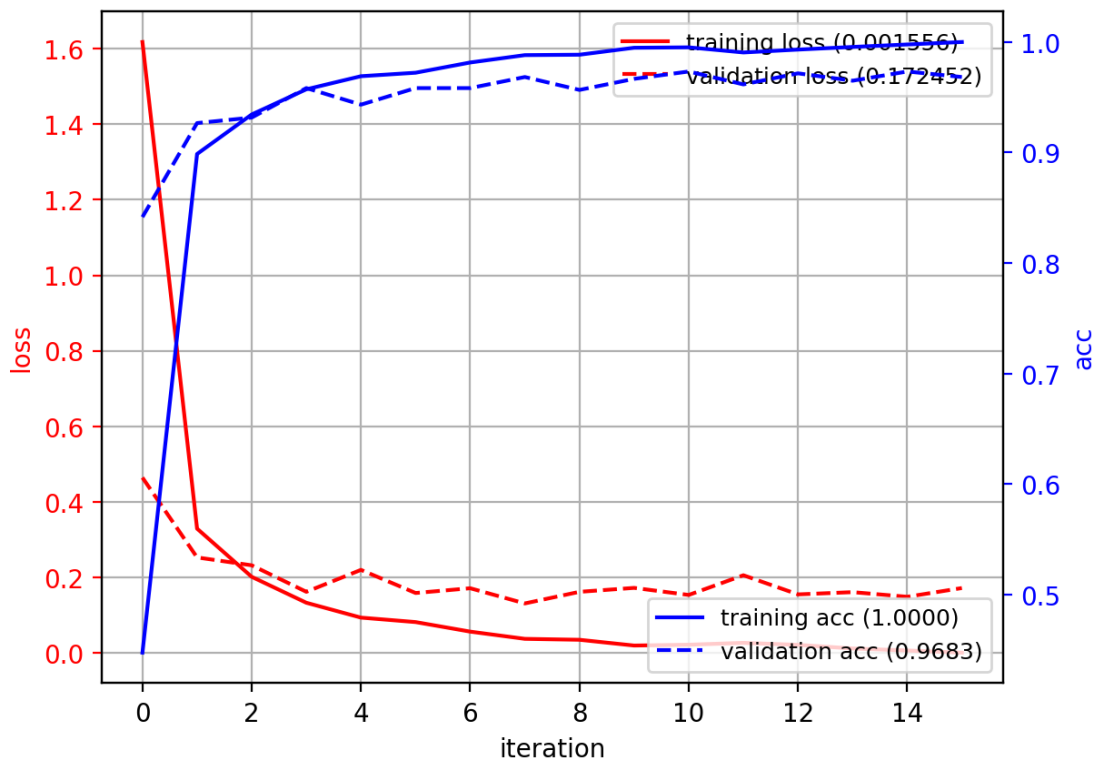
Total Trainable Parameters: 286940

Total Un-Trainable Parameters: 0

Using device: cuda

Epoch 1/100: Train Loss: 1.617102, Train Acc: 44.76%; Val Loss: 0.464670,
Val Acc: 84.17%

Epoch 2/100: Train Loss: 0.329649, Train Acc: 89.87%; Val Loss: 0.253477,
 Val Acc: 92.67%
 Epoch 3/100: Train Loss: 0.202198, Train Acc: 93.44%; Val Loss: 0.232633,
 Val Acc: 93.17%
 Epoch 4/100: Train Loss: 0.133558, Train Acc: 95.72%; Val Loss: 0.162527,
 Val Acc: 95.83%
 Epoch 5/100: Train Loss: 0.094230, Train Acc: 96.91%; Val Loss: 0.220510,
 Val Acc: 94.33%
 EarlyStopping counter: 1 out of 5
 Epoch 6/100: Train Loss: 0.082415, Train Acc: 97.22%; Val Loss: 0.159395,
 Val Acc: 95.83%
 EarlyStopping counter: 2 out of 5
 Epoch 7/100: Train Loss: 0.057166, Train Acc: 98.15%; Val Loss: 0.171946,
 Val Acc: 95.83%
 EarlyStopping counter: 3 out of 5
 Epoch 8/100: Train Loss: 0.038058, Train Acc: 98.81%; Val Loss: 0.131769,
 Val Acc: 96.83%
 Epoch 9/100: Train Loss: 0.035580, Train Acc: 98.85%; Val Loss: 0.162469,
 Val Acc: 95.67%
 EarlyStopping counter: 1 out of 5
 Epoch 10/100: Train Loss: 0.020369, Train Acc: 99.48%; Val Loss: 0.172894,
 Val Acc: 96.67%
 EarlyStopping counter: 2 out of 5
 Epoch 11/100: Train Loss: 0.022506, Train Acc: 99.52%; Val Loss: 0.154210,
 Val Acc: 97.33%
 Epoch 12/100: Train Loss: 0.027414, Train Acc: 99.06%; Val Loss: 0.206340,
 Val Acc: 96.17%
 EarlyStopping counter: 1 out of 5
 Epoch 13/100: Train Loss: 0.022148, Train Acc: 99.31%; Val Loss: 0.155686,
 Val Acc: 97.17%
 EarlyStopping counter: 2 out of 5
 Epoch 14/100: Train Loss: 0.012529, Train Acc: 99.56%; Val Loss: 0.161637,
 Val Acc: 96.50%
 EarlyStopping counter: 3 out of 5
 Epoch 15/100: Train Loss: 0.007216, Train Acc: 99.78%; Val Loss: 0.149385,
 Val Acc: 97.33%
 EarlyStopping counter: 4 out of 5
 Epoch 16/100: Train Loss: 0.001556, Train Acc: 100.00%; Val Loss: 0.172452,
 Val Acc: 96.83%
 EarlyStopping counter: 5 out of 5
 Early stopping triggered



```
[16]: test_dataset = TensorDataset(testI, torch.from_numpy(testY).long())
test_loader = DataLoader(test_dataset, batch_size=batch_size, shuffle=False)
acc = evaluate_model(model, test_loader)

print("test accuracy:", acc)
```

test accuracy: 96.7

11 Outline

- Going deeper
 - ReLU and Batchnorm
- **Optimization methods**
- Deep architectures and Image classification
- Transfer learning

12 Optimization with SGD

- Ideally, we would like to use all the samples to compute the gradient, but this is too time consuming.
- Use a *minibatch* (a few samples) at a time to estimate the gradient.
 - creates an unbiased estimator of the gradient.

- the variance (expected squared error) depends on the number of samples.
 - * i.e., the estimated gradient is *noisy*.

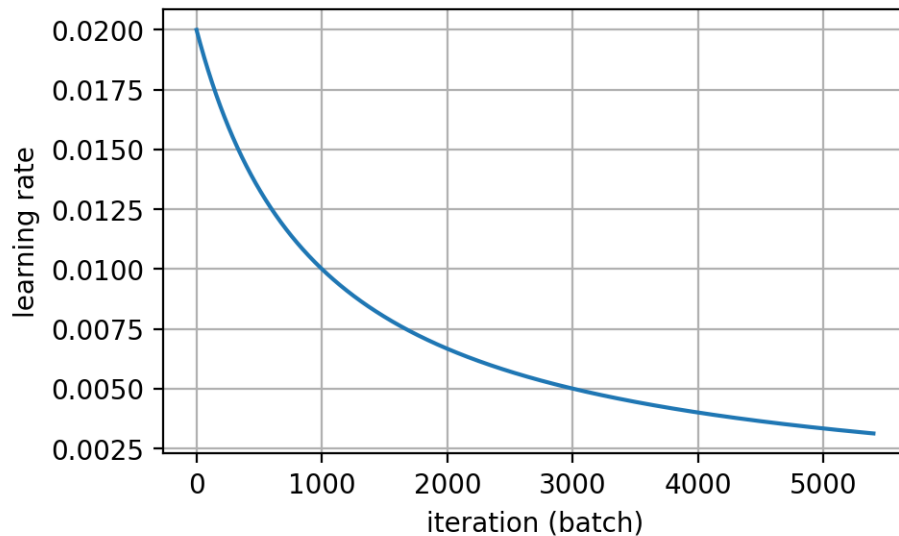
13 Learning rates

- **Problem:** For gradient descent, at the minimum we should have $\frac{dL}{d\mathbf{w}} = 0$.
- *What about for SGD?*
 - at the minimum $\frac{dL}{d\mathbf{w}} \neq 0$ because of noise in the gradient.
 - * SGD still moves around.
- **Solution:** reduce the learning rate during the epochs.
 - Examples: for iteration/epoch k ,
 - * **linear change:** $\eta_k = (1 - \alpha)\eta_0 + \alpha\eta_T$, where $\alpha = k/T$, and η_0, η_T given.
 - * **decay:** $\eta_k = \frac{1}{1+\delta k}\eta_0$, where $0 < \delta < 1$.
 - we want a small learning rate when we are close to the minimum.
 - * needs to be set empirically by examining the learning curves.

14 Example: Keras decay

- use the built-in `decay` parameter.
 - applied after each batch.

```
[17]: plt.figure(figsize=(5,3))
      its = arange(0,50*5400/50) # 50 epochs, 5400/50 iterations per epoch
      lr = 0.02*(1./(1+its*1e-3))
      plt.plot(its, lr)
      plt.grid(); plt.xlabel('iteration (batch)'); plt.ylabel('learning rate');
      plt.show()
```



```
[18]: # Create model instance
model = DeepCNN()

# Move model to device (GPU if available)
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
model = model.to(device)
print(f"Using device: {device}")

criterion = nn.CrossEntropyLoss()
optimizer = optim.SGD(model.parameters(), lr=0.02, momentum=0.9, nesterov=True)
scheduler = ExponentialLR(optimizer, gamma=math.pow(0.95, 1/108))

# Train the model
history, model = train_model(model, train_loader, valid_loader, criterion,
    ↪optimizer, epochs=50, patience=-1, batch_scheduler=scheduler)
```

Using device: cuda

Epoch 1/50:	Train Loss: 1.716768, Train Acc: 40.04%;	Val Loss: 0.419060, Val Acc: 88.17%
Epoch 2/50:	Train Loss: 0.325775, Train Acc: 89.72%;	Val Loss: 0.260520, Val Acc: 93.17%
Epoch 3/50:	Train Loss: 0.202823, Train Acc: 93.93%;	Val Loss: 0.179392, Val Acc: 93.50%
Epoch 4/50:	Train Loss: 0.137527, Train Acc: 95.85%;	Val Loss: 0.182461, Val Acc: 95.00%
Epoch 5/50:	Train Loss: 0.091641, Train Acc: 97.06%;	Val Loss: 0.180758, Val Acc: 95.50%
Epoch 6/50:	Train Loss: 0.069151, Train Acc: 97.81%;	Val Loss: 0.158545, Val Acc: 95.00%
Epoch 7/50:	Train Loss: 0.051072, Train Acc: 98.31%;	Val Loss: 0.152725, Val Acc: 96.17%
Epoch 8/50:	Train Loss: 0.030280, Train Acc: 99.13%;	Val Loss: 0.161747, Val Acc: 96.00%
Epoch 9/50:	Train Loss: 0.022639, Train Acc: 99.33%;	Val Loss: 0.173594, Val Acc: 96.33%
Epoch 10/50:	Train Loss: 0.015947, Train Acc: 99.56%;	Val Loss: 0.162648, Val Acc: 96.67%
Epoch 11/50:	Train Loss: 0.011119, Train Acc: 99.63%;	Val Loss: 0.170903, Val Acc: 96.67%
Epoch 12/50:	Train Loss: 0.008652, Train Acc: 99.74%;	Val Loss: 0.166767, Val Acc: 96.67%
Epoch 13/50:	Train Loss: 0.002905, Train Acc: 99.98%;	Val Loss: 0.181154, Val Acc: 96.67%
Epoch 14/50:	Train Loss: 0.001629, Train Acc: 99.98%;	Val Loss: 0.172741, Val Acc: 97.33%
Epoch 15/50:	Train Loss: 0.001207, Train Acc: 100.00%;	Val Loss: 0.186913, Val Acc: 96.83%

Epoch 16/50: Train Loss: 0.000863, Train Acc: 100.00%; Val Loss: 0.189911, Val Acc: 96.83%

Epoch 17/50: Train Loss: 0.000713, Train Acc: 100.00%; Val Loss: 0.192111, Val Acc: 97.00%

Epoch 18/50: Train Loss: 0.000625, Train Acc: 100.00%; Val Loss: 0.194995, Val Acc: 97.00%

Epoch 19/50: Train Loss: 0.000567, Train Acc: 100.00%; Val Loss: 0.199100, Val Acc: 96.83%

Epoch 20/50: Train Loss: 0.000514, Train Acc: 100.00%; Val Loss: 0.199754, Val Acc: 97.17%

Epoch 21/50: Train Loss: 0.000472, Train Acc: 100.00%; Val Loss: 0.202644, Val Acc: 97.00%

Epoch 22/50: Train Loss: 0.000440, Train Acc: 100.00%; Val Loss: 0.205207, Val Acc: 96.83%

Epoch 23/50: Train Loss: 0.000418, Train Acc: 100.00%; Val Loss: 0.205804, Val Acc: 96.83%

Epoch 24/50: Train Loss: 0.000397, Train Acc: 100.00%; Val Loss: 0.206984, Val Acc: 97.33%

Epoch 25/50: Train Loss: 0.000374, Train Acc: 100.00%; Val Loss: 0.209511, Val Acc: 97.00%

Epoch 26/50: Train Loss: 0.000354, Train Acc: 100.00%; Val Loss: 0.210647, Val Acc: 97.17%

Epoch 27/50: Train Loss: 0.000342, Train Acc: 100.00%; Val Loss: 0.211823, Val Acc: 97.00%

Epoch 28/50: Train Loss: 0.000330, Train Acc: 100.00%; Val Loss: 0.213149, Val Acc: 97.17%

Epoch 29/50: Train Loss: 0.000316, Train Acc: 100.00%; Val Loss: 0.214542, Val Acc: 97.17%

Epoch 30/50: Train Loss: 0.000307, Train Acc: 100.00%; Val Loss: 0.215214, Val Acc: 97.17%

Epoch 31/50: Train Loss: 0.000298, Train Acc: 100.00%; Val Loss: 0.215653, Val Acc: 97.17%

Epoch 32/50: Train Loss: 0.000289, Train Acc: 100.00%; Val Loss: 0.216484, Val Acc: 97.17%

Epoch 33/50: Train Loss: 0.000281, Train Acc: 100.00%; Val Loss: 0.217421, Val Acc: 97.17%

Epoch 34/50: Train Loss: 0.000275, Train Acc: 100.00%; Val Loss: 0.218286, Val Acc: 97.17%

Epoch 35/50: Train Loss: 0.000270, Train Acc: 100.00%; Val Loss: 0.218745, Val Acc: 97.17%

Epoch 36/50: Train Loss: 0.000264, Train Acc: 100.00%; Val Loss: 0.219660, Val Acc: 97.17%

Epoch 37/50: Train Loss: 0.000259, Train Acc: 100.00%; Val Loss: 0.220107, Val Acc: 97.00%

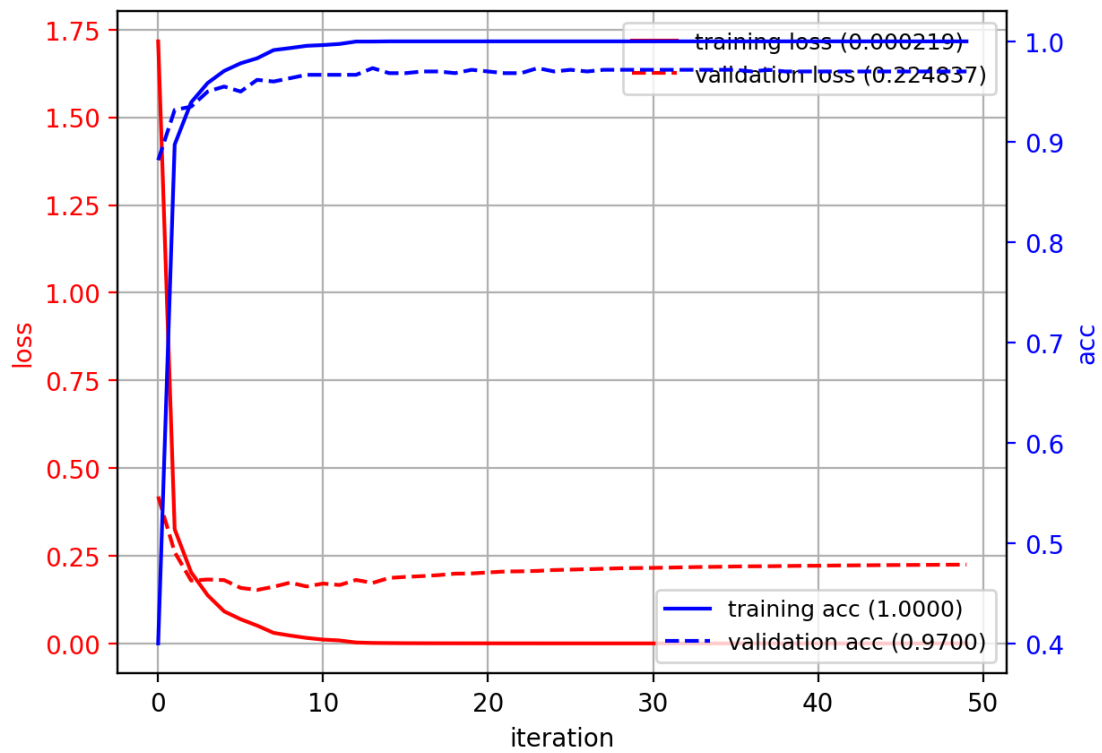
Epoch 38/50: Train Loss: 0.000254, Train Acc: 100.00%; Val Loss: 0.220290, Val Acc: 97.17%

Epoch 39/50: Train Loss: 0.000250, Train Acc: 100.00%; Val Loss: 0.220998, Val Acc: 97.00%

Epoch 40/50: Train Loss: 0.000246, Train Acc: 100.00%; Val Loss: 0.221544,
Val Acc: 97.00%
Epoch 41/50: Train Loss: 0.000243, Train Acc: 100.00%; Val Loss: 0.221706,
Val Acc: 97.00%
Epoch 42/50: Train Loss: 0.000239, Train Acc: 100.00%; Val Loss: 0.222382,
Val Acc: 97.00%
Epoch 43/50: Train Loss: 0.000235, Train Acc: 100.00%; Val Loss: 0.222725,
Val Acc: 97.00%
Epoch 44/50: Train Loss: 0.000232, Train Acc: 100.00%; Val Loss: 0.223023,
Val Acc: 97.00%
Epoch 45/50: Train Loss: 0.000230, Train Acc: 100.00%; Val Loss: 0.223515,
Val Acc: 97.00%
Epoch 46/50: Train Loss: 0.000228, Train Acc: 100.00%; Val Loss: 0.223785,
Val Acc: 97.00%
Epoch 47/50: Train Loss: 0.000225, Train Acc: 100.00%; Val Loss: 0.224014,
Val Acc: 97.00%
Epoch 48/50: Train Loss: 0.000223, Train Acc: 100.00%; Val Loss: 0.224324,
Val Acc: 97.00%
Epoch 49/50: Train Loss: 0.000221, Train Acc: 100.00%; Val Loss: 0.224666,
Val Acc: 97.00%
Epoch 50/50: Train Loss: 0.000219, Train Acc: 100.00%; Val Loss: 0.224837,
Val Acc: 97.00%

```
[19]: # Plot the training history
      plot_history(history)

      acc = evaluate_model(model, test_loader)
      print("test accuracy:", acc)
```



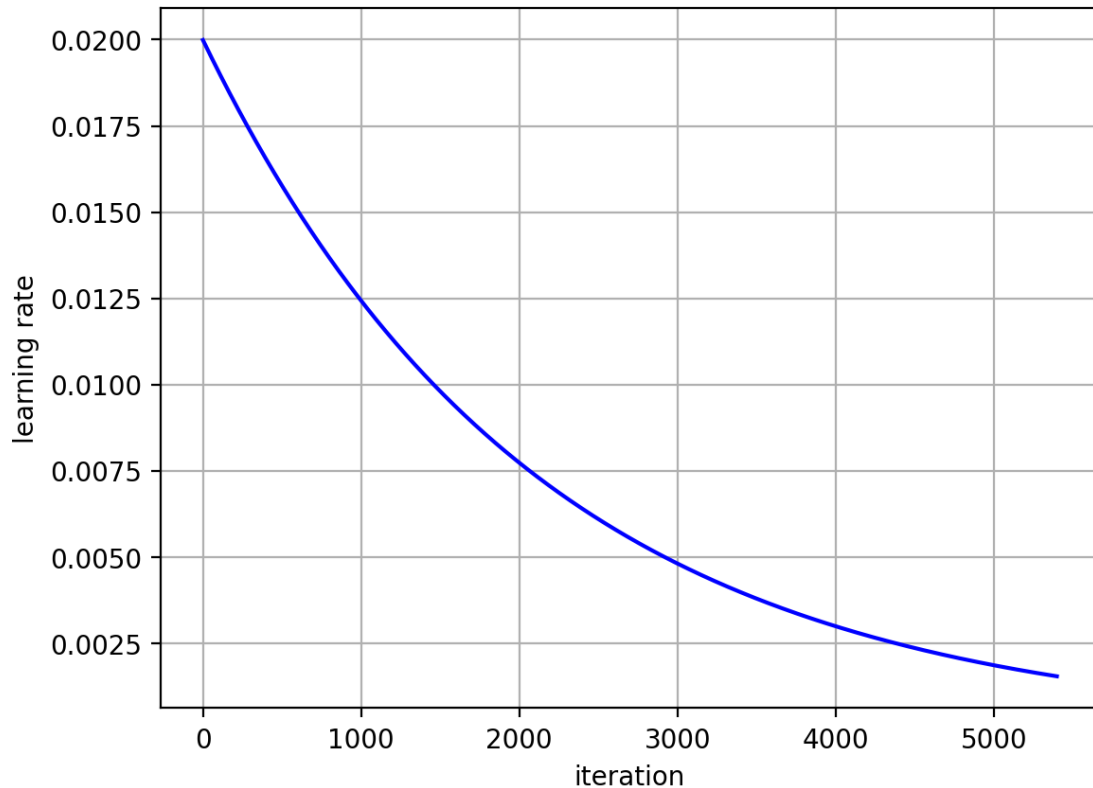
test accuracy: 96.62

```
[20]: def plot_lr(history):
    fig, ax1 = plt.subplots()

    ax1.plot(history['learning_rate'], 'b')
    ax1.grid(True)
    ax1.set_xlabel('iteration')
    ax1.set_ylabel('learning rate')

    plt.show()

plot_lr(history)
```

15 Adaptive schedule

- reduce the learning rate when the validation loss no longer improves
 - similar to early stopping criteria
- implemented as a callback function

```
[21]: # Create model instance
model = DeepCNN()

# Move model to device (GPU if available)
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
model = model.to(device)
print(f"Using device: {device}")

criterion = nn.CrossEntropyLoss() # This includes softmax + cross entropy
optimizer = optim.SGD(model.parameters(), lr=0.02, momentum=0.9, nesterov=True)
# reduce LR by a factor of 0.1, if no change in 5 epochs
scheduler = ReduceLROnPlateau(optimizer, mode='min', factor=0.1, patience=5)

# Train the model
```

```
history, model = train_model(model, train_loader, valid_loader, criterion, □
                               ↪optimizer, epochs=50, patience=-1, epoch_scheduler=scheduler)
```

Using device: cuda

Epoch 1/50:	Train Loss: 1.584269, Train Acc: 47.15%;	Val Loss: 0.404391, Val Acc: 87.17%
Epoch 2/50:	Train Loss: 0.313745, Train Acc: 90.24%;	Val Loss: 0.241543, Val Acc: 92.67%
Epoch 3/50:	Train Loss: 0.191844, Train Acc: 93.94%;	Val Loss: 0.236455, Val Acc: 92.83%
Epoch 4/50:	Train Loss: 0.130987, Train Acc: 95.72%;	Val Loss: 0.192679, Val Acc: 94.50%
Epoch 5/50:	Train Loss: 0.097266, Train Acc: 96.61%;	Val Loss: 0.177332, Val Acc: 94.83%
Epoch 6/50:	Train Loss: 0.072049, Train Acc: 97.61%;	Val Loss: 0.172696, Val Acc: 95.00%
Epoch 7/50:	Train Loss: 0.057187, Train Acc: 98.19%;	Val Loss: 0.157365, Val Acc: 95.67%
Epoch 8/50:	Train Loss: 0.051671, Train Acc: 98.46%;	Val Loss: 0.145465, Val Acc: 97.00%
Epoch 9/50:	Train Loss: 0.025971, Train Acc: 99.09%;	Val Loss: 0.254315, Val Acc: 94.50%
Epoch 10/50:	Train Loss: 0.023875, Train Acc: 99.20%;	Val Loss: 0.148523, Val Acc: 96.83%
Epoch 11/50:	Train Loss: 0.016409, Train Acc: 99.46%;	Val Loss: 0.193763, Val Acc: 96.00%
Epoch 12/50:	Train Loss: 0.024521, Train Acc: 99.19%;	Val Loss: 0.198806, Val Acc: 95.00%
Epoch 13/50:	Train Loss: 0.014728, Train Acc: 99.61%;	Val Loss: 0.173626, Val Acc: 97.17%
Epoch 14/50:	Train Loss: 0.005073, Train Acc: 99.87%;	Val Loss: 0.208359, Val Acc: 96.83%
Epoch 15/50:	Train Loss: 0.000861, Train Acc: 100.00%;	Val Loss: 0.207530, Val Acc: 96.83%
Epoch 16/50:	Train Loss: 0.000659, Train Acc: 100.00%;	Val Loss: 0.207451, Val Acc: 97.17%
Epoch 17/50:	Train Loss: 0.000588, Train Acc: 100.00%;	Val Loss: 0.207748, Val Acc: 97.17%
Epoch 18/50:	Train Loss: 0.000541, Train Acc: 100.00%;	Val Loss: 0.207970, Val Acc: 97.17%
Epoch 19/50:	Train Loss: 0.000506, Train Acc: 100.00%;	Val Loss: 0.208384, Val Acc: 97.17%
Epoch 20/50:	Train Loss: 0.000477, Train Acc: 100.00%;	Val Loss: 0.208821, Val Acc: 97.33%
Epoch 21/50:	Train Loss: 0.000454, Train Acc: 100.00%;	Val Loss: 0.208852, Val Acc: 97.33%
Epoch 22/50:	Train Loss: 0.000452, Train Acc: 100.00%;	Val Loss: 0.208900, Val Acc: 97.33%

Val Acc: 97.33%

Epoch 23/50: Train Loss: 0.000449, Train Acc: 100.00%; Val Loss: 0.208964,
Val Acc: 97.33%

Epoch 24/50: Train Loss: 0.000447, Train Acc: 100.00%; Val Loss: 0.209015,
Val Acc: 97.33%

Epoch 25/50: Train Loss: 0.000445, Train Acc: 100.00%; Val Loss: 0.209046,
Val Acc: 97.33%

Epoch 26/50: Train Loss: 0.000443, Train Acc: 100.00%; Val Loss: 0.209096,
Val Acc: 97.33%

Epoch 27/50: Train Loss: 0.000441, Train Acc: 100.00%; Val Loss: 0.209106,
Val Acc: 97.33%

Epoch 28/50: Train Loss: 0.000441, Train Acc: 100.00%; Val Loss: 0.209115,
Val Acc: 97.33%

Epoch 29/50: Train Loss: 0.000441, Train Acc: 100.00%; Val Loss: 0.209125,
Val Acc: 97.33%

Epoch 30/50: Train Loss: 0.000441, Train Acc: 100.00%; Val Loss: 0.209133,
Val Acc: 97.33%

Epoch 31/50: Train Loss: 0.000440, Train Acc: 100.00%; Val Loss: 0.209137,
Val Acc: 97.33%

Epoch 32/50: Train Loss: 0.000440, Train Acc: 100.00%; Val Loss: 0.209143,
Val Acc: 97.33%

Epoch 33/50: Train Loss: 0.000440, Train Acc: 100.00%; Val Loss: 0.209143,
Val Acc: 97.33%

Epoch 34/50: Train Loss: 0.000440, Train Acc: 100.00%; Val Loss: 0.209143,
Val Acc: 97.33%

Epoch 35/50: Train Loss: 0.000440, Train Acc: 100.00%; Val Loss: 0.209143,
Val Acc: 97.33%

Epoch 36/50: Train Loss: 0.000440, Train Acc: 100.00%; Val Loss: 0.209144,
Val Acc: 97.33%

Epoch 37/50: Train Loss: 0.000440, Train Acc: 100.00%; Val Loss: 0.209145,
Val Acc: 97.33%

Epoch 38/50: Train Loss: 0.000440, Train Acc: 100.00%; Val Loss: 0.209145,
Val Acc: 97.33%

Epoch 39/50: Train Loss: 0.000440, Train Acc: 100.00%; Val Loss: 0.209145,
Val Acc: 97.33%

Epoch 40/50: Train Loss: 0.000440, Train Acc: 100.00%; Val Loss: 0.209145,
Val Acc: 97.33%

Epoch 41/50: Train Loss: 0.000440, Train Acc: 100.00%; Val Loss: 0.209145,
Val Acc: 97.33%

Epoch 42/50: Train Loss: 0.000440, Train Acc: 100.00%; Val Loss: 0.209145,
Val Acc: 97.33%

Epoch 43/50: Train Loss: 0.000440, Train Acc: 100.00%; Val Loss: 0.209145,
Val Acc: 97.33%

Epoch 44/50: Train Loss: 0.000440, Train Acc: 100.00%; Val Loss: 0.209145,
Val Acc: 97.33%

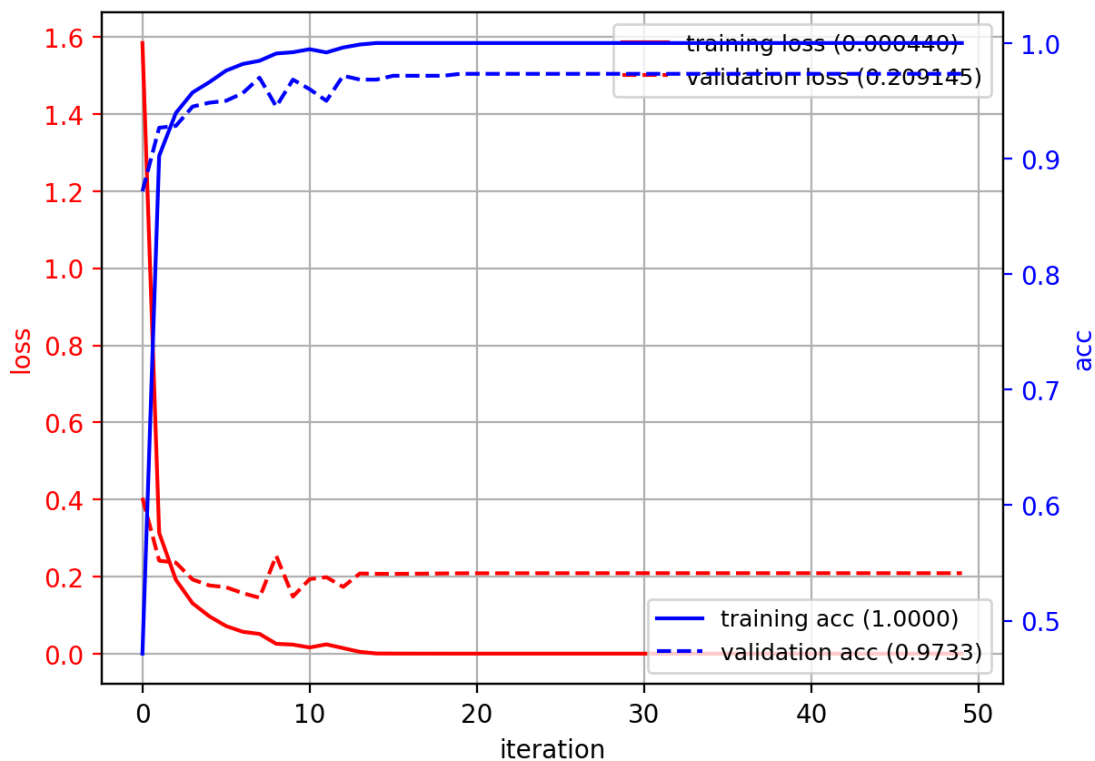
Epoch 45/50: Train Loss: 0.000440, Train Acc: 100.00%; Val Loss: 0.209145,
Val Acc: 97.33%

Epoch 46/50: Train Loss: 0.000440, Train Acc: 100.00%; Val Loss: 0.209145,

Val Acc: 97.33%
 Epoch 47/50: Train Loss: 0.000440, Train Acc: 100.00%; Val Loss: 0.209145, Val Acc: 97.33%
 Epoch 48/50: Train Loss: 0.000440, Train Acc: 100.00%; Val Loss: 0.209145, Val Acc: 97.33%
 Epoch 49/50: Train Loss: 0.000440, Train Acc: 100.00%; Val Loss: 0.209145, Val Acc: 97.33%
 Epoch 50/50: Train Loss: 0.000440, Train Acc: 100.00%; Val Loss: 0.209145, Val Acc: 97.33%

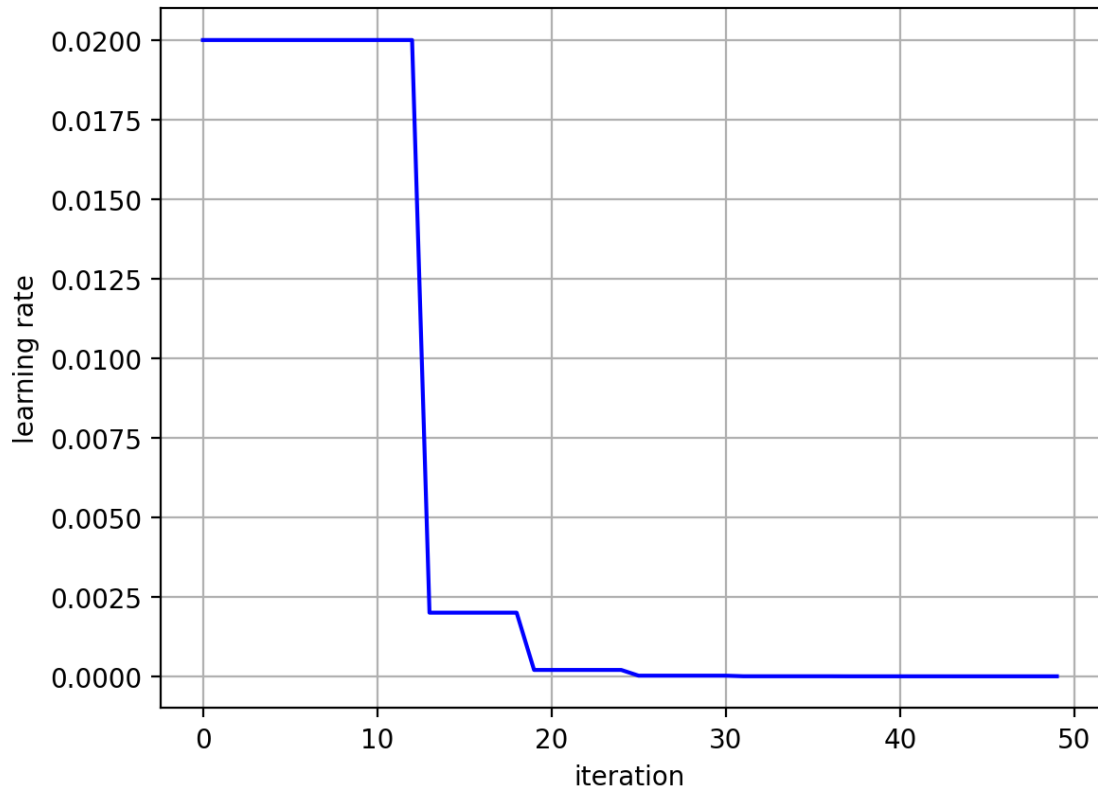
```
[22]: # Plot the training history
plot_history(history)

acc = evaluate_model(model, test_loader)
print("test accuracy:", acc)
```



test accuracy: 96.42

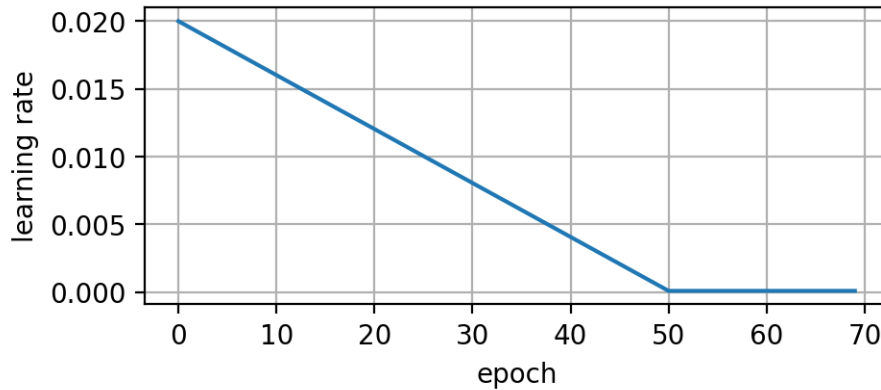
```
[23]: plot_lr(history)
```



16 Fixed schedule

- specify our own schedule using callback `LearningRateScheduler`
- pass a schedule function
- inputs are the epoch and current learning rate.
- outputs the learning rate for this epoch.

```
[24]: def sc(epoch, curlr):
        alpha = minimum(epoch/50, 1.)
        return 0.02*(1-alpha)+0.0001*alpha
epoch = arange(0,70)
plt.figure(figsize=(5,2))
plt.plot(epoch, sc(epoch,None))
plt.grid(); plt.xlabel('epoch'); plt.ylabel('learning rate');plt.show()
```



```
[25]: # Create model instance
model = DeepCNN()

# Move model to device (GPU if available)
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
model = model.to(device)
print(f"Using device: {device}")

criterion = nn.CrossEntropyLoss() # This includes softmax + cross entropy
optimizer = optim.SGD(model.parameters(), lr=0.02, momentum=0.9, nesterov=True)
# reduce LR by a factor of 0.1, if no change in 5 epochs
scheduler = LinearLR(optimizer, start_factor=1.0, end_factor=0.05,
    ↪total_iters=50)

# Train the model
history, model = train_model(model, train_loader, valid_loader, criterion,
    ↪optimizer, epochs=50, patience=-1, epoch_scheduler=scheduler)
```

Using device: cuda

Epoch 1/50:	Train Loss: 1.561114, Train Acc: 45.91%;	Val Loss: 0.400893, Val Acc: 87.00%
Epoch 2/50:	Train Loss: 0.305708, Train Acc: 90.65%;	Val Loss: 0.233811, Val Acc: 92.83%
Epoch 3/50:	Train Loss: 0.183516, Train Acc: 94.56%;	Val Loss: 0.199219, Val Acc: 94.33%
Epoch 4/50:	Train Loss: 0.128340, Train Acc: 95.87%;	Val Loss: 0.186400, Val Acc: 94.83%
Epoch 5/50:	Train Loss: 0.085602, Train Acc: 97.31%;	Val Loss: 0.175606, Val Acc: 96.00%
Epoch 6/50:	Train Loss: 0.068440, Train Acc: 97.72%;	Val Loss: 0.208534, Val Acc: 95.17%
Epoch 7/50:	Train Loss: 0.053286, Train Acc: 98.31%;	Val Loss: 0.150136, Val Acc: 97.17%

Epoch 8/50: Train Loss: 0.033672, Train Acc: 99.04%; Val Loss: 0.166403, Val Acc: 95.33%

Epoch 9/50: Train Loss: 0.034847, Train Acc: 98.89%; Val Loss: 0.157852, Val Acc: 96.17%

Epoch 10/50: Train Loss: 0.019101, Train Acc: 99.43%; Val Loss: 0.181433, Val Acc: 96.67%

Epoch 11/50: Train Loss: 0.021414, Train Acc: 99.28%; Val Loss: 0.172300, Val Acc: 97.33%

Epoch 12/50: Train Loss: 0.006556, Train Acc: 99.85%; Val Loss: 0.205704, Val Acc: 96.83%

Epoch 13/50: Train Loss: 0.006376, Train Acc: 99.80%; Val Loss: 0.172653, Val Acc: 97.67%

Epoch 14/50: Train Loss: 0.001764, Train Acc: 99.96%; Val Loss: 0.188386, Val Acc: 97.33%

Epoch 15/50: Train Loss: 0.000815, Train Acc: 100.00%; Val Loss: 0.193944, Val Acc: 97.33%

Epoch 16/50: Train Loss: 0.000452, Train Acc: 100.00%; Val Loss: 0.198022, Val Acc: 97.33%

Epoch 17/50: Train Loss: 0.000335, Train Acc: 100.00%; Val Loss: 0.201504, Val Acc: 97.67%

Epoch 18/50: Train Loss: 0.000283, Train Acc: 100.00%; Val Loss: 0.206944, Val Acc: 97.50%

Epoch 19/50: Train Loss: 0.000247, Train Acc: 100.00%; Val Loss: 0.210983, Val Acc: 97.50%

Epoch 20/50: Train Loss: 0.000224, Train Acc: 100.00%; Val Loss: 0.212869, Val Acc: 97.50%

Epoch 21/50: Train Loss: 0.000201, Train Acc: 100.00%; Val Loss: 0.214884, Val Acc: 97.50%

Epoch 22/50: Train Loss: 0.000186, Train Acc: 100.00%; Val Loss: 0.218093, Val Acc: 97.50%

Epoch 23/50: Train Loss: 0.000173, Train Acc: 100.00%; Val Loss: 0.219498, Val Acc: 97.67%

Epoch 24/50: Train Loss: 0.000162, Train Acc: 100.00%; Val Loss: 0.220969, Val Acc: 97.67%

Epoch 25/50: Train Loss: 0.000153, Train Acc: 100.00%; Val Loss: 0.222589, Val Acc: 97.67%

Epoch 26/50: Train Loss: 0.000143, Train Acc: 100.00%; Val Loss: 0.224058, Val Acc: 97.67%

Epoch 27/50: Train Loss: 0.000137, Train Acc: 100.00%; Val Loss: 0.225604, Val Acc: 97.67%

Epoch 28/50: Train Loss: 0.000130, Train Acc: 100.00%; Val Loss: 0.226975, Val Acc: 97.67%

Epoch 29/50: Train Loss: 0.000125, Train Acc: 100.00%; Val Loss: 0.227996, Val Acc: 97.67%

Epoch 30/50: Train Loss: 0.000121, Train Acc: 100.00%; Val Loss: 0.228760, Val Acc: 97.67%

Epoch 31/50: Train Loss: 0.000116, Train Acc: 100.00%; Val Loss: 0.229694, Val Acc: 97.67%

```

Epoch 32/50:   Train Loss: 0.000112, Train Acc: 100.00%;   Val Loss: 0.230692,
Val Acc: 97.67%
Epoch 33/50:   Train Loss: 0.000109, Train Acc: 100.00%;   Val Loss: 0.231587,
Val Acc: 97.67%
Epoch 34/50:   Train Loss: 0.000106, Train Acc: 100.00%;   Val Loss: 0.232389,
Val Acc: 97.67%
Epoch 35/50:   Train Loss: 0.000103, Train Acc: 100.00%;   Val Loss: 0.233086,
Val Acc: 97.67%
Epoch 36/50:   Train Loss: 0.000100, Train Acc: 100.00%;   Val Loss: 0.233730,
Val Acc: 97.67%
Epoch 37/50:   Train Loss: 0.000098, Train Acc: 100.00%;   Val Loss: 0.234420,
Val Acc: 97.67%
Epoch 38/50:   Train Loss: 0.000096, Train Acc: 100.00%;   Val Loss: 0.234842,
Val Acc: 97.67%
Epoch 39/50:   Train Loss: 0.000094, Train Acc: 100.00%;   Val Loss: 0.235350,
Val Acc: 97.67%
Epoch 40/50:   Train Loss: 0.000092, Train Acc: 100.00%;   Val Loss: 0.235751,
Val Acc: 97.67%
Epoch 41/50:   Train Loss: 0.000091, Train Acc: 100.00%;   Val Loss: 0.236222,
Val Acc: 97.67%
Epoch 42/50:   Train Loss: 0.000090, Train Acc: 100.00%;   Val Loss: 0.236660,
Val Acc: 97.67%
Epoch 43/50:   Train Loss: 0.000088, Train Acc: 100.00%;   Val Loss: 0.236924,
Val Acc: 97.67%
Epoch 44/50:   Train Loss: 0.000087, Train Acc: 100.00%;   Val Loss: 0.237250,
Val Acc: 97.67%
Epoch 45/50:   Train Loss: 0.000086, Train Acc: 100.00%;   Val Loss: 0.237504,
Val Acc: 97.67%
Epoch 46/50:   Train Loss: 0.000086, Train Acc: 100.00%;   Val Loss: 0.237803,
Val Acc: 97.67%
Epoch 47/50:   Train Loss: 0.000085, Train Acc: 100.00%;   Val Loss: 0.237978,
Val Acc: 97.67%
Epoch 48/50:   Train Loss: 0.000084, Train Acc: 100.00%;   Val Loss: 0.238225,
Val Acc: 97.67%
Epoch 49/50:   Train Loss: 0.000084, Train Acc: 100.00%;   Val Loss: 0.238314,
Val Acc: 97.67%
Epoch 50/50:   Train Loss: 0.000083, Train Acc: 100.00%;   Val Loss: 0.238428,
Val Acc: 97.67%

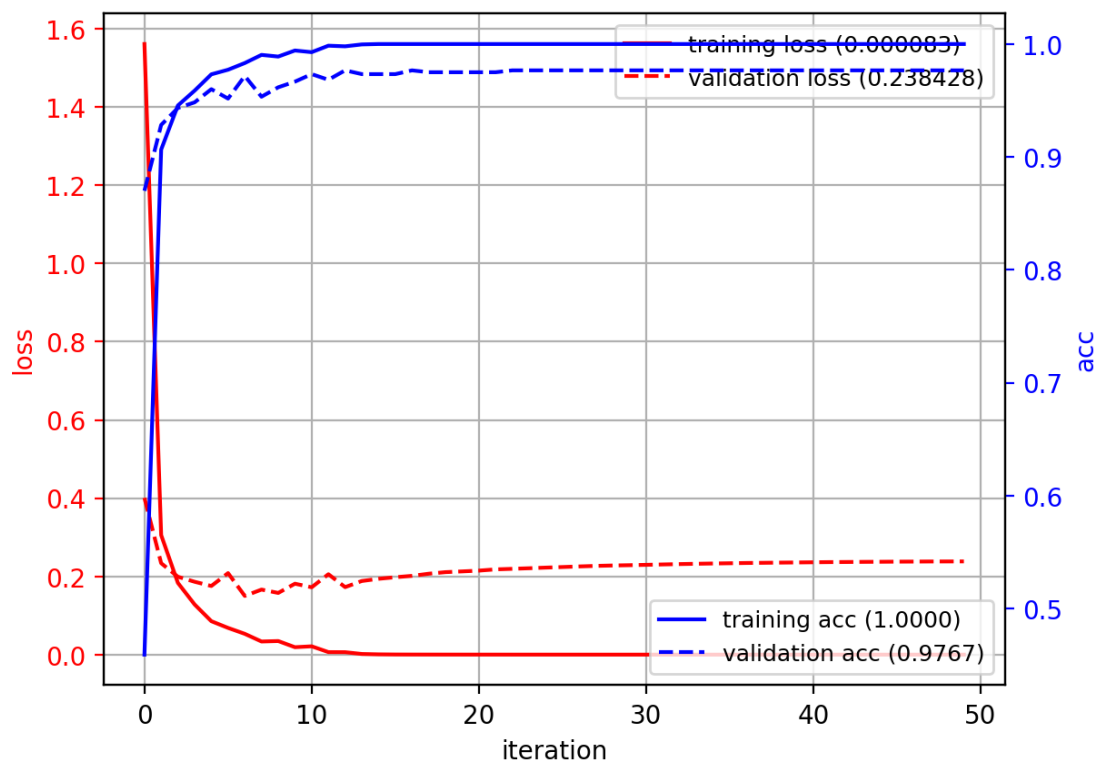
```

```

[26]: # Plot the training history
      plot_history(history)

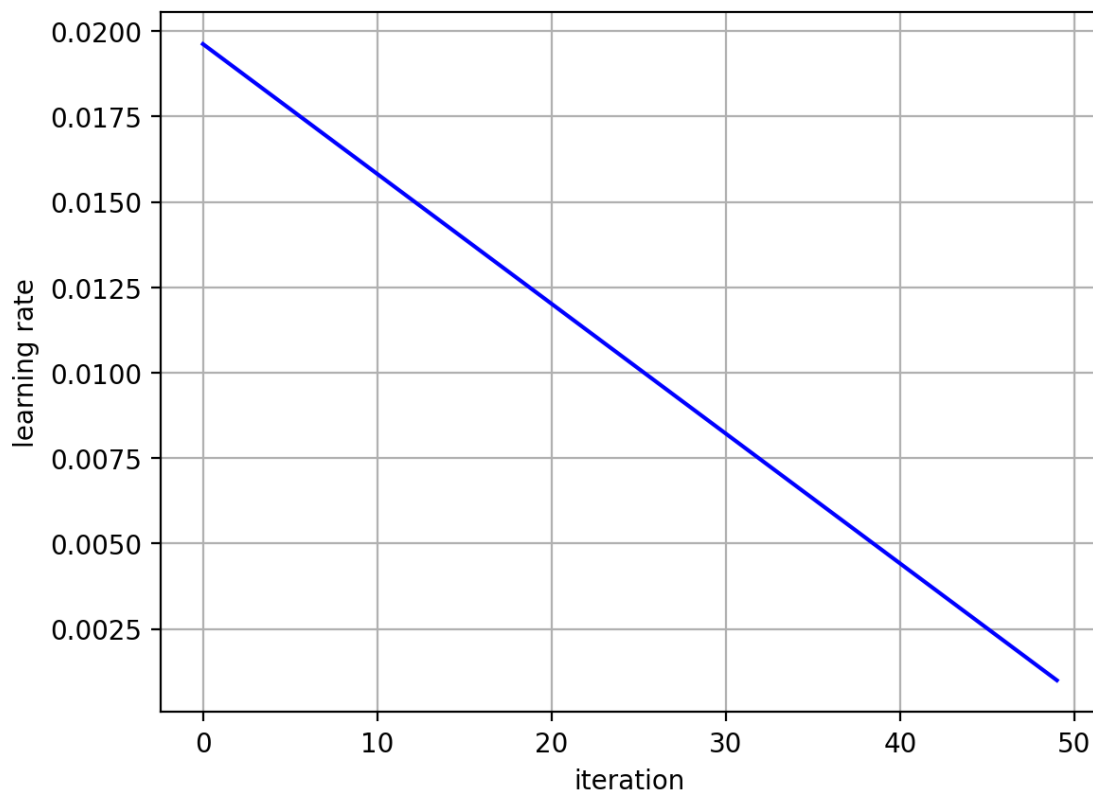
      acc = evaluate_model(model, test_loader)
      print("test accuracy:", acc)

```

test accuracy: 96.46

```
[27]: plot_lr(history)
```



17 Momentum

- **Problem:** The estimated gradient is noisy, can jump around.
- **Solution:** keep a running average of the gradients across mini-batches.
 - velocity: $\mathbf{v}^{(t)} = \alpha \mathbf{v}^{(t-1)} - \eta \left. \frac{dL}{d\mathbf{w}} \right|_{\mathbf{w}^{(t-1)}}$
 - * accumulate the gradients
 - * α is the momentum hyperparameter; how much it exponentially decays.
 - parameter update: $\mathbf{w}^{(t)} \leftarrow \mathbf{w}^{(t-1)} + \mathbf{v}^{(t)}$
- Example:
 - red path is using momentum
 - black arrows show the gradient directions at each step
 - without momentum, the path would oscillate wildly.

18 Nesterov Momentum

- Compute the gradient *after* the current velocity is applied.
 - interim update: $\tilde{\mathbf{w}} = \mathbf{w}^{(t-1)} + \alpha \mathbf{v}^{(t-1)}$
 - velocity: $\mathbf{v}^{(t)} = \alpha \mathbf{v}^{(t-1)} - \eta \left. \frac{dL}{d\mathbf{w}} \right|_{\tilde{\mathbf{w}}}$
 - parameter update: $\mathbf{w}^{(t)} \leftarrow \mathbf{w}^{(t-1)} + \mathbf{v}^{(t)}$
- Adds a *correction factor* to improve convergence (for convex batch case)

19 Why does SGD work?

- The loss function has many local minima
- SGD adds “noise” to the true gradient.
 - the noise allows escaping/avoiding/jumping over small local minima.
- Example:
 - red arrow = true gradient
 - green arrow = added noise
 - black arrow = computed gradient

20 SGD smoothes the loss function

- the added gradient noise is equivalent to convolving the loss function with the noise density.
- higher learning rate $\rightarrow$ larger noise $\rightarrow$ smoother loss
- smoother loss removes the local minimum, making it easier to get near the global minimum.
 - but not exactly on it.
- need to reduce the learning rate in stages to converge to the global optimum.

21 Optimization with Adaptive Learning Rates

- Introduce a separate learning rate for each parameter, and automatically adapt the learning rates during optimization.
- **AdaGrad** (`torch.optim.Adagrad`)
 - adapt individual learning rates by dividing by the square-root of the gradient energy accumulated over the iterations.
 - * $\mathbf{g} = \frac{dL}{d\mathbf{w}}$
 - * $\mathbf{r}^{(t)} = \mathbf{r}^{(t-1)} + \mathbf{g}^2$
 - * $\mathbf{w}^{(t)} = \mathbf{w}^{(t-1)} - \frac{\eta}{\delta + \sqrt{\mathbf{r}}} \mathbf{g}$
 - (operations are element-wise)
- **RMSProp** (`torch.optim.RMSprop`)
 - use exponential decay on the accumulated energy:
 - * $\mathbf{r}^{(t)} = \rho \mathbf{r}^{(t-1)} + (1 - \rho) \mathbf{g}^2$
- **Adam** (`torch.optim.Adam`)
 - use momentum with exponential weighting to estimate the gradient and gradient energy.
 - * $\mathbf{s}^{(t)} = \rho_1 \mathbf{s}^{(t-1)} + (1 - \rho_1) \mathbf{g}$
 - * $\mathbf{r}^{(t)} = \rho_2 \mathbf{r}^{(t-1)} + (1 - \rho_2) \mathbf{g}^2$
 - adds a bias correction for these two estimates.
 - * $\hat{\mathbf{s}}^{(t)} = \frac{1}{1 - \rho_1^t} \mathbf{s}^{(t)}$
 - * $\hat{\mathbf{r}}^{(t)} = \frac{1}{1 - \rho_2^t} \mathbf{r}^{(t)}$
 - update:
 - * $\mathbf{w}^{(t)} = \mathbf{w}^{(t-1)} - \frac{\eta}{\delta + \sqrt{\hat{\mathbf{r}}^{(t)}}} \hat{\mathbf{s}}^{(t)}$

22 Example

- change the optimizer when compiling the network.

```
[28]: # Create model instance
model = DeepCNN()

# Move model to device (GPU if available)
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
model = model.to(device)
print(f"Using device: {device}")

criterion = nn.CrossEntropyLoss()
optimizer = optim.Adam(model.parameters(), lr=0.001)

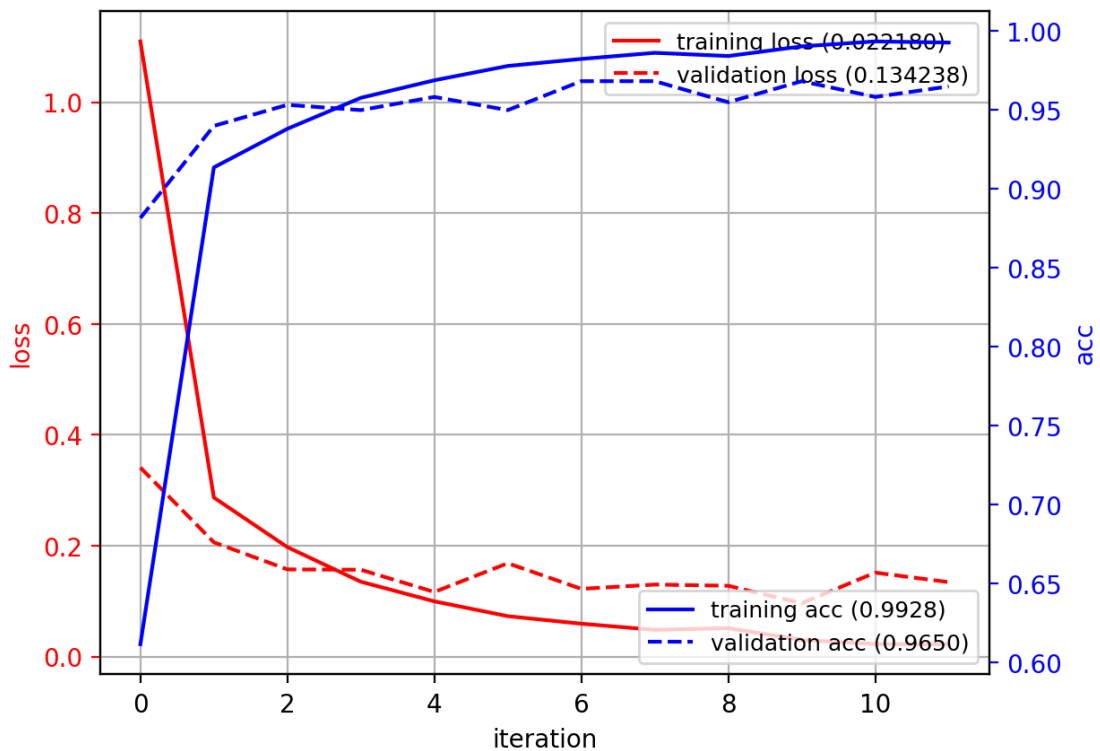
# Train the model
history, model = train_model(model, train_loader, valid_loader, criterion,
                               optimizer, epochs=100, patience=5)
```

```
Using device: cuda
Epoch 1/100:   Train Loss: 1.109883, Train Acc: 61.17%;   Val Loss: 0.340950,
Val Acc: 88.17%
Epoch 2/100:   Train Loss: 0.286864, Train Acc: 91.37%;   Val Loss: 0.206138,
Val Acc: 94.00%
Epoch 3/100:   Train Loss: 0.197446, Train Acc: 93.81%;   Val Loss: 0.157247,
Val Acc: 95.33%
Epoch 4/100:   Train Loss: 0.134986, Train Acc: 95.78%;   Val Loss: 0.156642,
Val Acc: 95.00%
EarlyStopping counter: 1 out of 5
Epoch 5/100:   Train Loss: 0.099617, Train Acc: 96.89%;   Val Loss: 0.116509,
Val Acc: 95.83%
Epoch 6/100:   Train Loss: 0.072853, Train Acc: 97.80%;   Val Loss: 0.168416,
Val Acc: 95.00%
EarlyStopping counter: 1 out of 5
Epoch 7/100:   Train Loss: 0.059275, Train Acc: 98.24%;   Val Loss: 0.122134,
Val Acc: 96.83%
Epoch 8/100:   Train Loss: 0.048134, Train Acc: 98.63%;   Val Loss: 0.129905,
Val Acc: 96.83%
EarlyStopping counter: 1 out of 5
Epoch 9/100:   Train Loss: 0.051222, Train Acc: 98.43%;   Val Loss: 0.127682,
Val Acc: 95.50%
EarlyStopping counter: 2 out of 5
Epoch 10/100:  Train Loss: 0.029914, Train Acc: 99.04%;   Val Loss: 0.096202,
Val Acc: 96.83%
EarlyStopping counter: 3 out of 5
Epoch 11/100:  Train Loss: 0.022572, Train Acc: 99.35%;   Val Loss: 0.151418,
Val Acc: 95.83%
EarlyStopping counter: 4 out of 5
```

Epoch 12/100: Train Loss: 0.022180, Train Acc: 99.28%; Val Loss: 0.134238, Val Acc: 96.50%
EarlyStopping counter: 5 out of 5
Early stopping triggered

```
[29]: # Plot the training history
plot_history(history)

acc = evaluate_model(model, test_loader)
print("test accuracy:", acc)
```



test accuracy: 96.52

23 Which optimizer is best?

- there's no best optimizer...
- based on the problem and own familiarity with tuning the hyperparameters.

Lecture7b

October 27, 2025

1 CS5489 - Machine Learning

2 Lecture 7b - Deep Learning

2.0.1 Dept. of Computer Science, City University of Hong Kong

3 Outline

- Going deeper
 - ReLU and Batchnorm
- Optimization methods
- **Deep architectures and Image classification**
- Transfer learning

```
[1]: # setup
%matplotlib inline
import matplotlib_inline # setup output image format
matplotlib_inline.backend_inline.set_matplotlib_formats('retina')
import matplotlib.pyplot as plt
plt.rcParams['figure.dpi'] = 100 # display larger images
import matplotlib
from numpy import *
import numpy as np
from sklearn import *
from scipy import stats
from scipy import signal
import random
import math
from prettytable import PrettyTable

rbow = plt.get_cmap('rainbow')
```

```
[2]: import torch
import torchvision
import torch.nn as nn
import torch.optim as optim
from torch.utils.data import DataLoader, TensorDataset
from torchvision import transforms
```

```

from torchvision.transforms import ToTensor
from torchvision.transforms import functional as F
import struct
import sys
print(f"Python: {sys.version} PyTorch: {torch.__version__}")

# Set device
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
print(f"Using device: {device}")

```

Python: 3.13.5 | packaged by Anaconda, Inc. | (main, Jun 12 2025, 16:09:02) [GCC 11.2.0] PyTorch: 2.8.0+cu129
Using device: cuda

```

[3]: def plot_history(history):
    fig, ax1 = plt.subplots()

    ax1.plot(history['train_loss'], 'r', label="training loss {:.6f}").
    ↪format(history['train_loss'][-1]))
    ax1.plot(history['val_loss'], 'r--', label="validation loss {:.6f}").
    ↪format(history['val_loss'][-1]))
    ax1.grid(True)
    ax1.set_xlabel('iteration')
    ax1.legend(loc="best", fontsize=9)
    ax1.set_ylabel('loss', color='r')
    ax1.tick_params('y', colors='r')

    if 'train_accuracy' in history:
        ax2 = ax1.twinx()

        ax2.plot(history['train_accuracy'], 'b', label="training acc {:.4f}").
        ↪format(history['train_accuracy'][-1]))
        ax2.plot(history['val_accuracy'], 'b--', label="validation acc {:.
        ↪4f}").format(history['val_accuracy'][-1]))

        ax2.legend(loc="best", fontsize=9)
        ax2.set_ylabel('acc', color='b')
        ax2.tick_params('y', colors='b')
    plt.show()

```

```

[4]: def show_imgs(W_list, nc=10, highlight_green=None, highlight_red=None,
    ↪titles=None):
    nfilter = len(W_list)
    nr = (nfilter - 1) // nc + 1
    for i in range(nr):
        for j in range(nc):
            idx = i * nc + j

```

```

    if idx == nfilter:
        break
    plt.subplot(nr, nc, idx + 1)
    cur_W = W_list[idx]
    if ndim(cur_W) == 2:
        plt.imshow(cur_W, cmap='gray', interpolation='nearest')
    else:
        plt.imshow(cur_W, interpolation='nearest')

    if titles is not None:
        if isinstance(titles, str):
            plt.title(titles % idx)
        elif isinstance(titles, list):
            plt.title(titles[idx])
        else:
            plt.title(titles)

    if ((highlight_green is not None) and highlight_green[idx]) or \
        ((highlight_red is not None) and highlight_red[idx]):
        ax = plt.gca()
        if highlight_green[idx]:
            mycol = '#00FF00'
        else:
            mycol = 'r'
        for S in ['bottom', 'top', 'right', 'left']:
            ax.spines[S].set_color(mycol)
            ax.spines[S].set_lw(2.0)
        ax.xaxis.set_ticks_position('none')
        ax.yaxis.set_ticks_position('none')
        ax.set_xticks([])
        ax.set_yticks([])
    else:
        plt.gca().set_axis_off()

plt.show()

```

4 Image Classification

- **Goal:** given an image, predict the object class
 - typically only one main object is present
 - If enough classes are considered, then it's a generic high-level vision task

5 ImageNet

- ImageNet Large Scale Visual Recognition Challenge (ILSVRC)
 - 1000 image classes
 - 1.2 million images

- Human performance: ~5.1% (top-5 classification error)

6 Performance of Deep Learning

- The introduction of ILSVRC coincided with the emergence of Deep Learning.
- Top-5 error rate decreased as deeper NN were developed
 - Not just deeper, but also the architecture design was smarter

7 LeNet-5 (1998)

- The standard CNN architecture
 - 7 layers
 - convolutions & pooling, final fully-connected layer
- Designed for hand-written digit recognition (MNIST)

8 AlexNet

- Similar architecture to LeNet, but deeper (14 layers)
 - 11x11, 5x5, 3x3 convolutions
- Tricks used: DropOut, ReLU, max pooling, data augmentation, SGD momentum
- One of the first networks trained on GPU
 - (it is split in two pipelines because it was trained on 2 GPUs simultaneously)

9 VGGnet

- Same style as LeNet and AlexNet
- Design choices:
 - only use 3x3 convolution filters (less parameters)
 - stack several convolution layers one after another, followed by pooling
 - * equivalent to using a larger filter, but with less parameters.
 - number of feature channels doubles after each stage.
 - * more higher-level features

10 Progress so far...

- Deeper...more parameters...better accuracy...
 - *performance is saturating*

11 InceptionNet

- “Network-in-Network” architecture
 - Core building blocks (modules), consisting of several layers, are combined repeatedly.
- Inception Module
 - several convolution filters in parallel
 - * extract features at different scales (1x1, 3x3, 5x5)
 - * pool features (3x3 max)

- features are concatenated and passed to next block.

12 InceptionNet (V1)

- 9 inception modules, 22 layers
 - 50 convolution blocks
- Auxiliary classification tasks
 - using features in the middle of the network to perform classification (yellow boxes)

13 Auxiliary tasks

- **Goal:** strengthen the supervisory signal to the early layers
- perform the same classification task with branches off the main network.
- new loss is sum of individual classification losses:
 - $L = \lambda_0 L_0 + \lambda_1 L_1 + \lambda_2 L_2$
 - $\frac{dL}{d\mathbf{w}} = \lambda_0 \frac{dL_0}{d\mathbf{w}} + \lambda_1 \frac{dL_1}{d\mathbf{w}} + \lambda_2 \frac{dL_2}{d\mathbf{w}}$
- auxiliary tasks shorten the path to compute the gradients of early layers.
 - more direct supervision to the earlier layers.
 - encourages earlier layers to be more discriminative.

14 Multi-task learning

- several related tasks are learned together using a shared base network.
- combined loss: $L = \lambda_0 L_0 + \lambda_1 L_1 + \lambda_2 L_2$
- multiple supervisory signals make the common network find good features for all tasks.
 - gradients are accumulated in the common base network.
- if the tasks are related, the features for one task could be useful for the other tasks.
 - e.g., facial landmark locations could help face recognition.

15 Going deeper

- **Idea:** when stacking more layers, we should not have degraded performance, since we could make the new layer as an “identity” layer.
 - if $\mathbf{A} = \mathbf{I}$, then $f_i(\mathbf{x}) = \mathbf{A}^T \mathbf{x} = \mathbf{x}$
 - i.e., this layer doesn’t change the hidden values from the previous layer.
- **Problem:** if we use L2 regularization on $\mathbf{A}$, then we encourage $\mathbf{A} \rightarrow 0$.
 - so we cannot get this behavior...

16 Residual Learning

- **Solution:** learn a residual function so that if $\mathbf{A} = 0$, then $f(\mathbf{x}) = \mathbf{x}$
 - residual function $F(\mathbf{x}) = \mathbf{A}^T \mathbf{x}$
 - output: $f(\mathbf{x}) = F(\mathbf{x}) + \mathbf{x}$
 - * learn the $\mathbf{A}$ in the residual.

- * when $\mathbf{A} = 0$, then $f(\mathbf{x}) = \mathbf{x}$.
- the “identity path” is sometimes called a “shortcut” connection.

17 Residual Blocks

- Different versions of the blocks.
 - differences in where to add the identity
 - (e) is preferred.

18 Stacking Residual Blocks

- The network is learning a function (image to class)
 - build the function by appending residual blocks
 - each block learns a residual, which is added to the previous block
 - * keep all the previous information and make small changes with the residual

19 Residual Network (ResNet)

- 34 layers, 50 layers, 100 layers, 1000 layers
 - 3x3 filters
 - residual connection every two layers.

20 ResNet as an Ensemble

- Consider a 3-block ResNet
- Look at the equations:

$$\begin{aligned} \mathbf{z}_1 &= \mathbf{x} + f_1(\mathbf{x}), \\ \mathbf{z}_2 &= \mathbf{z}_1 + f_2(\mathbf{z}_1), \\ \mathbf{z}_3 &= \mathbf{z}_2 + f_3(\mathbf{z}_2) \end{aligned}$$

- Substitute for $\mathbf{z}_2$...

$$\mathbf{z}_3 = \underbrace{\mathbf{z}_1 + f_2(\mathbf{z}_1)}_{\text{blue}} + f_3(\underbrace{\mathbf{z}_1 + f_2(\mathbf{z}_1)}_{\text{blue}})$$

- Substitute for $\mathbf{z}_1$...

$$\mathbf{z}_3 = \underbrace{\mathbf{x} + f_1(\mathbf{x})}_{\text{red}} + f_2(\underbrace{\mathbf{x} + f_1(\mathbf{x})}_{\text{red}}) + f_3(\underbrace{\mathbf{x} + f_1(\mathbf{x})}_{\text{red}} + \underbrace{f_2(\mathbf{x} + f_1(\mathbf{x}))}_{\text{red}})$$

- the output is sum of different combinations of residual blocks

$$\mathbf{z}_3 = \underbrace{\mathbf{x} + f_1(\mathbf{x})}_{\text{red}} + \underbrace{f_2(\mathbf{x} + f_1(\mathbf{x}))}_{\text{orange}} + \underbrace{f_3(\mathbf{x} + f_1(\mathbf{x})) + f_2(\mathbf{x} + f_1(\mathbf{x}))}_{\text{orange}}$$

- equivalent to an *ensemble*!

- *How many paths are there?*

- can select block f_i or not, thus 2^N paths.
 - * where N is the number of residual blocks.
- 2^N models in the ensemble

21 Robustness

- Because it is an ensemble, ResNet is robust to removal of a few blocks (sub-models).
 - error is still small when a few blocks are removed.
 - can be exploited to make prunable NN by ordering the blocks according to their importance.

22 Paths and Gradients

- (a) the paths through the ResNet has different lengths
 - number of paths with length $L = “N \text{ choose } L”$, $N = \text{number of blocks}$.
- (b) shorter paths have more effective gradients because they have direct supervision from the loss.
 - reduces the vanishing gradient problem.
- (c) although there are many paths, the short paths contain most of the gradient signal.
 - effectively, the network is shallow during training.

23 DenseNet

- Each layer gets inputs from all previous layers
 - Since feature maps are passed forward, fewer output channels are required in each layer.
 - last layer uses all previous feature maps (of various semantic levels).

24 Attention Mechanisms

- ignore unimportant features (channels or spatial regions) in the feature map by using a (soft) mask.
- i.e., attend to only important features (or spatial regions)
- “Squeeze-Excitation Block”
 - “squeeze” feature map into 1×1 channel vector
 - “excite” features by scaling up important ones, and scaling down less-important ones.

25 SEnet

- the SE block can be incorporated into standard architectures.

26 Error vs. Computation vs. Number of Parameters

- Just adding more layers is limited.

- Too many parameters
- More improvement by changing the architecture design.
 - Smarter use of the parameters
 - **Architecture Design is important!**

27 Moving forwards

- Performance with CNN-style architectures seems to be saturated.
- **Advantage:**
 - inductive bias (assumptions) of CNNs: translation equivariance, locality
- **Disadvantages:**
 - cannot well model long-range dependencies (global context)
 - * in the receptive field, 2 far-apart pixels are connected via a long path.
 - low speed - adding more layers cannot be parallelized.
- **Solution:** need a new type of architecture.

28 Transformers

- NLP sequence-to-sequence model (e.g. machine translation)
 - remove previous inductive biases: no recurrences, no convolutions
 - processes all words (inputs) in parallel
 - selectively attend to different locations with a *self-attention* mechanism

29 Encoder

- Stack of 6 blocks with identical architecture
 - input & output: sequence of word tokens
 - multi-head self-attention (MHA): find correlated features
 - residual connection: MHA updates the current features
 - layer normalization: normalize all features within layer to $N(0, 1)$
 - position-wise feed-forward NN (MLP): identical for each position.
 - Input tokens $X \in \mathbb{R}^{D \times n}$ for a batch of n tokens.

30 Dot-Product Attention

- Attention mechanism has 3 parts:
 - n is the number of input words (positions).
 - query $\mathbf{Q} = W^Q X = [\mathbf{q}_1 \dots \mathbf{q}_n] \in \mathbb{R}^{d_k \times n}$: what we want to match.
 - key $\mathbf{K} = W^K X = [\mathbf{k}_1 \dots \mathbf{k}_n] \in \mathbb{R}^{d_k \times n}$: what we match against.
 - value $\mathbf{V} = W^V X = [\mathbf{v}_1 \dots \mathbf{v}_n] \in \mathbb{R}^{d_v \times n}$: what to output when there is a match.
 - $W^Q, W^K \in \mathbb{R}^{d_k \times D}$, $W^V \in \mathbb{R}^{d_v \times D}$, where d_k, d_v are the dimensions of the key-query space and value space and D is the embedding (model) dimension ($D = 12288$ for GPT-3).
- Attention matrix:
 - $\mathbf{A} = [\mathbf{a}_1 \dots \mathbf{a}_n] = \text{softmax}(\frac{1}{\sqrt{d_k}} \mathbf{K}^T \mathbf{Q}) \in \mathbb{R}^{n \times n}$
 - * $\mathbf{a}_j = \text{softmax}(\frac{1}{\sqrt{d_k}} \mathbf{K}^T \mathbf{q}_j)$ is the vector of matches for the j -th query (to all keys).

* A_{ij} is the match between the j -th query and the i -th key.

- Output: $\text{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \mathbf{V}\mathbf{A} = [\mathbf{V}\mathbf{a}_1 \cdots \mathbf{V}\mathbf{a}_n]$
 - for the j -th query, linear combination of values for matched keys.

31 Multi-Head Attention

- Input: $X \in \mathbb{R}^{D \times n}$
- Perform multiple dot-product attentions in parallel
- Each *head* projects the input X to its own lower-dim key/query/value space that is learned
 - head weights: $\mathbf{W}_i^Q, \mathbf{W}_i^K \in \mathbb{R}^{d_k \times D}, \mathbf{W}_i^V \in \mathbb{R}^{d_v \times D}$
 - head QKV: $\mathbf{Q}' = \mathbf{W}_i^Q X, \mathbf{K}' = \mathbf{W}_i^K X, \mathbf{V}' = \mathbf{W}_i^V X$
 - head output: $\mathbf{H}_i = \text{Attn}(\mathbf{Q}', \mathbf{K}', \mathbf{V}')$
- Output: concatenate head outputs and project to original space
 - $\text{MHA}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \mathbf{W}^O \begin{bmatrix} \mathbf{H}_1 \\ \vdots \\ \mathbf{H}_h \end{bmatrix}$
 - $\mathbf{W}^O \in \mathbb{R}^{D \times h d_v}$
 - We set the output dim of $\mathbf{W}^O$ to be D since we hope to recover the original embedding dimension.
 - This means that after concatenation $d_v h = D$.
- Example dimensions:
 - number of heads $h = 8$, embedding dimension $D = 512$
 - $d_k = d_v = D/h = 64$

32 Other things

- position-wise NN (2-layer MLP) applies to each token position at end of each block.
 - weights shared for all token positions; different for each layer.
- learnable input embeddings: turn each word (token) into a vector.
 - $\mathbf{x}' = \mathbf{A}\mathbf{x}$
 - * $\mathbf{x}$ is a one-hot encoding of the word.
 - * $\mathbf{A}$ is a matrix of embedding vectors.
- position-encoding
 - the relative positions between inputs is not built into the self-attention architecture (c.f., convolution layer)
 - add a position-embedding to each word vector at the input.
 - learnable or fixed (sine and cosine waves of different frequencies)

32.1 Decoder

- adds mask to MHA to predict the current word only using previous words.
- apply MHA on output of the encoder.

33 Vision Transformers (ViT)

- Apply transformer architecture to image classification

1. Split image into patches (16x16 pixels).
 2. Flatten and then embed with linear projection.
 3. Prepend a learnable “[class]” token.
- Apply transformer architecture to image classification
 4. Add position embeddings
 5. Apply transformer encoder
 6. Classification head: MLP on the first output of the encoder.

34 Model variations

- Different variants with different depth.
- Can also apply ViT to the feature map from a CNN (Hybrid model)

35 Model Pre-training & Results

- Pre-training on different datasets, test on ImageNet1k
 - worse than ResNets (BiT) when trained on IN1k (1k classes, 1.3M images)
 - similar when trained on IN21k (21k classes, 14M images)
 - better when trained on JFT (18k classes, 303M images)
 - **Why?**
- **Reason:**
 - The inductive biases in ResNet helps on smaller datasets (IN1k), but ViT does not have enough data to learn this.
 - Large scale training on JFT, then ViT can outperform the inductive bias.

36 Visualization & Interpretation

- Visualization of the models attention, linear embeddings, position encoding, and attention head

</center>

37 Error vs. Computation vs. Number of Parameters

- Transformer and self-attention mechanism
 - no clean theory (yet), overall observation is that more data + larger model = better generalization.
 - give rise to a series of empirical conjectures called “Scaling Laws”.
 - A notable one is Chichilla scaling law (Hoffmann et al, 2022): $\mathcal{L}(N, T) = L_0 + AT^{-\alpha} + BN^{-\beta}$, where N, T are model size and data size respectively. Other constants can be learned to fit different LLMs.
 - Ideologically similar to “Moore’s Law”, so this isn’t a real law.

38 Image Classifiers

- Keras includes several pre-trained image classifier networks

- VGG16, VGG19, ResNet50, DenseNet, InceptionV3, ...
- already trained on ImageNet
- Can use these networks to:
 - perform image classification (for 1000 classes only)
 - extract image features for our own task
 - transfer learning - modify/adapt a pre-trained network for our own task
- Import the ResNet model class and support functions
- Load and pre-process the image for the network

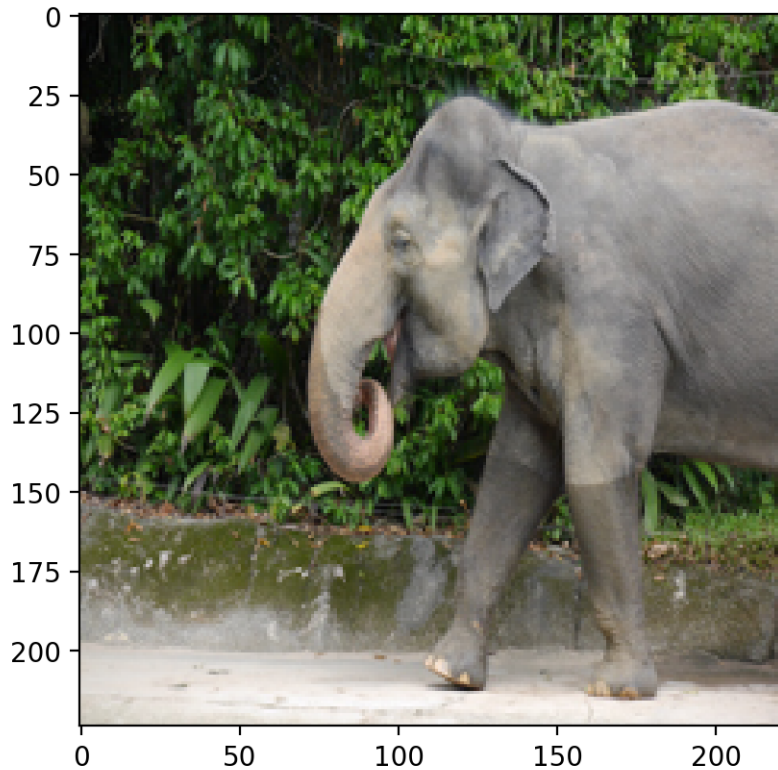
```
[5]: # contains the model and support functions for ResNet50
from PIL import Image
import torchvision.models as models

# load an image
img_path = 'imgs/elephant1.jpg'
img = Image.open(img_path).convert('RGB').resize((224,224)) # ResNet expects
↳ this size
plt.imshow(np.asarray(img))
plt.show()

# create an instance of the model
model = models.resnet50(weights=models.ResNet50_Weights.IMAGENET1K_V1) # you
↳ might use "model = models.resnet50(pretrained=True)" for older version
model.eval() # Set to evaluation mode

preprocess = transforms.Compose([
    transforms.Resize(256),
    transforms.CenterCrop(224),
    transforms.ToTensor(),
    transforms.Normalize(mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]),
])

xi = preprocess(img)
```

- Create an instance of the ResNet50 model
 - trained on ImageNet
 - will download the model if loading for the first time.
- Look at the 1000 classes

```
[6]: from torchvision.models import ResNet50_Weights
# Get the class names (equivalent to decode_predictions)
weights = ResNet50_Weights.IMAGENET1K_V1
class_names = weights.meta["categories"]
print("All ImageNet classes:")
print(class_names)
```

All ImageNet classes:

```
['tench', 'goldfish', 'great white shark', 'tiger shark', 'hammerhead',
'electric ray', 'stingray', 'cock', 'hen', 'ostrich', 'brambling', 'goldfinch',
'house finch', 'junco', 'indigo bunting', 'robin', 'bulbul', 'jay', 'magpie',
'chickadee', 'water ouzel', 'kite', 'bald eagle', 'vulture', 'great grey owl',
'European fire salamander', 'common newt', 'eft', 'spotted salamander',
'axolotl', 'bullfrog', 'tree frog', 'tailed frog', 'loggerhead', 'leatherback
turtle', 'mud turtle', 'terrapien', 'box turtle', 'banded gecko', 'common
iguana', 'American chameleon', 'whiptail', 'agama', 'frilled lizard', 'alligator
lizard', 'Gila monster', 'green lizard', 'African chameleon', 'Komodo dragon',
```

'African crocodile', 'American alligator', 'triceratops', 'thunder snake',
 'ringneck snake', 'hognose snake', 'green snake', 'king snake', 'garter snake',
 'water snake', 'vine snake', 'night snake', 'boa constrictor', 'rock python',
 'Indian cobra', 'green mamba', 'sea snake', 'horned viper', 'diamondback',
 'sidewinder', 'trilobite', 'harvestman', 'scorpion', 'black and gold garden
 spider', 'barn spider', 'garden spider', 'black widow', 'tarantula', 'wolf
 spider', 'tick', 'centipede', 'black grouse', 'ptarmigan', 'ruffed grouse',
 'prairie chicken', 'peacock', 'quail', 'partridge', 'African grey', 'macaw',
 'sulphur-crested cockatoo', 'lorikeet', 'coucal', 'bee eater', 'hornbill',
 'hummingbird', 'jacamar', 'toucan', 'drake', 'red-breasted merganser', 'goose',
 'black swan', 'tusker', 'echidna', 'platypus', 'wallaby', 'koala', 'wombat',
 'jellyfish', 'sea anemone', 'brain coral', 'flatworm', 'nematode', 'conch',
 'snail', 'slug', 'sea slug', 'chiton', 'chambered nautilus', 'Dungeness crab',
 'rock crab', 'fiddler crab', 'king crab', 'American lobster', 'spiny lobster',
 'crayfish', 'hermit crab', 'isopod', 'white stork', 'black stork', 'spoonbill',
 'flamingo', 'little blue heron', 'American egret', 'bittern', 'crane bird',
 'limpkin', 'European gallinule', 'American coot', 'bustard', 'ruddy turnstone',
 'red-backed sandpiper', 'redshank', 'dowitcher', 'oystercatcher', 'pelican',
 'king penguin', 'albatross', 'grey whale', 'killer whale', 'dugong', 'sea lion',
 'Chihuahua', 'Japanese spaniel', 'Maltese dog', 'Pekinese', 'Shih-Tzu',
 'Blenheim spaniel', 'papillon', 'toy terrier', 'Rhodesian ridgeback', 'Afghan
 hound', 'basset', 'beagle', 'bloodhound', 'bluetick', 'black-and-tan coonhound',
 'Walker hound', 'English foxhound', 'redbone', 'borzoi', 'Irish wolfhound',
 'Italian greyhound', 'whippet', 'Ibizan hound', 'Norwegian elkhound',
 'otterhound', 'Saluki', 'Scottish deerhound', 'Weimaraner', 'Staffordshire
 bullterrier', 'American Staffordshire terrier', 'Bedlington terrier', 'Border
 terrier', 'Kerry blue terrier', 'Irish terrier', 'Norfolk terrier', 'Norwich
 terrier', 'Yorkshire terrier', 'wire-haired fox terrier', 'Lakeland terrier',
 'Sealyham terrier', 'Airedale', 'cairn', 'Australian terrier', 'Dandie Dinmont',
 'Boston bull', 'miniature schnauzer', 'giant schnauzer', 'standard schnauzer',
 'Scotch terrier', 'Tibetan terrier', 'silky terrier', 'soft-coated wheaten
 terrier', 'West Highland white terrier', 'Lhasa', 'flat-coated retriever',
 'curly-coated retriever', 'golden retriever', 'Labrador retriever', 'Chesapeake
 Bay retriever', 'German short-haired pointer', 'vizsla', 'English setter',
 'Irish setter', 'Gordon setter', 'Brittany spaniel', 'clumber', 'English
 springer', 'Welsh springer spaniel', 'cocker spaniel', 'Sussex spaniel', 'Irish
 water spaniel', 'kuvasz', 'schipperke', 'groenendael', 'malinois', 'briard',
 'kelpie', 'komondor', 'Old English sheepdog', 'Shetland sheepdog', 'collie',
 'Border collie', 'Bouvier des Flandres', 'Rottweiler', 'German shepherd',
 'Doberman', 'miniature pinscher', 'Greater Swiss Mountain dog', 'Bernese
 mountain dog', 'Appenzeller', 'EntleBucher', 'boxer', 'bull mastiff', 'Tibetan
 mastiff', 'French bulldog', 'Great Dane', 'Saint Bernard', 'Eskimo dog',
 'malamute', 'Siberian husky', 'dalmatian', 'affenpinscher', 'basenji', 'pug',
 'Leonberg', 'Newfoundland', 'Great Pyrenees', 'Samoyed', 'Pomeranian', 'chow',
 'keeshond', 'Brabancon griffon', 'Pembroke', 'Cardigan', 'toy poodle',
 'miniature poodle', 'standard poodle', 'Mexican hairless', 'timber wolf', 'white
 wolf', 'red wolf', 'coyote', 'dingo', 'dhole', 'African hunting dog', 'hyena',
 'red fox', 'kit fox', 'Arctic fox', 'grey fox', 'tabby', 'tiger cat', 'Persian

cat', 'Siamese cat', 'Egyptian cat', 'cougar', 'lynx', 'leopard', 'snow leopard', 'jaguar', 'lion', 'tiger', 'cheetah', 'brown bear', 'American black bear', 'ice bear', 'sloth bear', 'mongoose', 'meerkat', 'tiger beetle', 'ladybug', 'ground beetle', 'long-horned beetle', 'leaf beetle', 'dung beetle', 'rhinoceros beetle', 'weevil', 'fly', 'bee', 'ant', 'grasshopper', 'cricket', 'walking stick', 'cockroach', 'mantis', 'cicada', 'leafhopper', 'lacewing', 'dragonfly', 'damselfly', 'admiral', 'ringlet', 'monarch', 'cabbage butterfly', 'sulphur butterfly', 'lycaenid', 'starfish', 'sea urchin', 'sea cucumber', 'wood rabbit', 'hare', 'Angora', 'hamster', 'porcupine', 'fox squirrel', 'marmot', 'beaver', 'guinea pig', 'sorrel', 'zebra', 'hog', 'wild boar', 'warthog', 'hippopotamus', 'ox', 'water buffalo', 'bison', 'ram', 'bighorn', 'ibex', 'hartebeest', 'impala', 'gazelle', 'Arabian camel', 'llama', 'weasel', 'mink', 'polecat', 'black-footed ferret', 'otter', 'skunk', 'badger', 'armadillo', 'three-toed sloth', 'orangutan', 'gorilla', 'chimpanzee', 'gibbon', 'siamang', 'guenon', 'patas', 'baboon', 'macaque', 'langur', 'colobus', 'proboscis monkey', 'marmoset', 'capuchin', 'howler monkey', 'titi', 'spider monkey', 'squirrel monkey', 'Madagascar cat', 'indri', 'Indian elephant', 'African elephant', 'lesser panda', 'giant panda', 'barracouta', 'eel', 'coho', 'rock beauty', 'anemone fish', 'sturgeon', 'gar', 'lionfish', 'puffer', 'abacus', 'abaya', 'academic gown', 'accordion', 'acoustic guitar', 'aircraft carrier', 'airliner', 'airship', 'altar', 'ambulance', 'amphibian', 'analog clock', 'apiary', 'apron', 'ashcan', 'assault rifle', 'backpack', 'bakery', 'balance beam', 'balloon', 'ballpoint', 'Band Aid', 'banjo', 'bannister', 'barbell', 'barber chair', 'barbershop', 'barn', 'barometer', 'barrel', 'barrow', 'baseball', 'basketball', 'bassinet', 'bassoon', 'bathing cap', 'bath towel', 'bathtub', 'beach wagon', 'beacon', 'beaker', 'bearskin', 'beer bottle', 'beer glass', 'bell cote', 'bib', 'bicycle-built-for-two', 'bikini', 'binder', 'binoculars', 'birdhouse', 'boathouse', 'bobsled', 'bolo tie', 'bonnet', 'bookcase', 'bookshop', 'bottlecap', 'bow', 'bow tie', 'brass', 'brassiere', 'breakwater', 'breastplate', 'broom', 'bucket', 'buckle', 'bulletproof vest', 'bullet train', 'butcher shop', 'cab', 'caldron', 'candle', 'cannon', 'canoe', 'can opener', 'cardigan', 'car mirror', 'carousel', 'carpenter's kit', 'carton', 'car wheel', 'cash machine', 'cassette', 'cassette player', 'castle', 'catamaran', 'CD player', 'cello', 'cellular telephone', 'chain', 'chainlink fence', 'chain mail', 'chain saw', 'chest', 'chiffonier', 'chime', 'china cabinet', 'Christmas stocking', 'church', 'cinema', 'cleaver', 'cliff dwelling', 'cloak', 'clog', 'cocktail shaker', 'coffee mug', 'coffeepot', 'coil', 'combination lock', 'computer keyboard', 'confectionery', 'container ship', 'convertible', 'corkscrew', 'cornet', 'cowboy boot', 'cowboy hat', 'cradle', 'crane', 'crash helmet', 'crate', 'crib', 'Crock Pot', 'croquet ball', 'crutch', 'cuirass', 'dam', 'desk', 'desktop computer', 'dial telephone', 'diaper', 'digital clock', 'digital watch', 'dining table', 'dishrag', 'dishwasher', 'disk brake', 'dock', 'dogsled', 'dome', 'doormat', 'drilling platform', 'drum', 'drumstick', 'dumbbell', 'Dutch oven', 'electric fan', 'electric guitar', 'electric locomotive', 'entertainment center', 'envelope', 'espresso maker', 'face powder', 'feather boa', 'file', 'fireboat', 'fire engine', 'fire screen', 'flagpole', 'flute', 'folding chair', 'football helmet', 'forklift', 'fountain', 'fountain pen', 'four-poster', 'freight car', 'French horn', 'frying pan', 'fur

coat', 'garbage truck', 'gasmask', 'gas pump', 'goblet', 'go-kart', 'golf ball',
'golfcart', 'gondola', 'gong', 'gown', 'grand piano', 'greenhouse', 'grille',
'grocery store', 'guillotine', 'hair slide', 'hair spray', 'half track',
'hammer', 'hamper', 'hand blower', 'hand-held computer', 'handkerchief', 'hard
disc', 'harmonica', 'harp', 'harvester', 'hatchet', 'holster', 'home theater',
'honeycomb', 'hook', 'hoopskirt', 'horizontal bar', 'horse cart', 'hourglass',
'iPod', 'iron', 'jack-o'-lantern', 'jean', 'jeep', 'jersey', 'jigsaw puzzle',
'jinrikisha', 'joystick', 'kimono', 'knee pad', 'knot', 'lab coat', 'ladle',
'lampshade', 'laptop', 'lawn mower', 'lens cap', 'letter opener', 'library',
'lifeboat', 'lighter', 'limousine', 'liner', 'lipstick', 'Loafer', 'lotion',
'loudspeaker', 'loupe', 'lumbermill', 'magnetic compass', 'mailbag', 'mailbox',
'maillot', 'maillot tank suit', 'manhole cover', 'maraca', 'marimba', 'mask',
'matchstick', 'maypole', 'maze', 'measuring cup', 'medicine chest', 'megalith',
'microphone', 'microwave', 'military uniform', 'milk can', 'minibus',
'miniskirt', 'minivan', 'missile', 'mitten', 'mixing bowl', 'mobile home',
'Model T', 'modem', 'monastery', 'monitor', 'moped', 'mortar', 'mortarboard',
'mosque', 'mosquito net', 'motor scooter', 'mountain bike', 'mountain tent',
'mouse', 'mousetrap', 'moving van', 'muzzle', 'nail', 'neck brace', 'necklace',
'nipple', 'notebook', 'obelisk', 'oboe', 'ocarina', 'odometer', 'oil filter',
'organ', 'oscilloscope', 'overskirt', 'oxcart', 'oxygen mask', 'packet',
'paddle', 'paddlewheel', 'padlock', 'paintbrush', 'pajama', 'palace', 'panpipe',
'paper towel', 'parachute', 'parallel bars', 'park bench', 'parking meter',
'passenger car', 'patio', 'pay-phone', 'pedestal', 'pencil box', 'pencil
sharpener', 'perfume', 'Petri dish', 'photocopier', 'pick', 'pickelhaube',
'picket fence', 'pickup', 'pier', 'piggy bank', 'pill bottle', 'pillow', 'ping-
pong ball', 'pinwheel', 'pirate', 'pitcher', 'plane', 'planetarium', 'plastic
bag', 'plate rack', 'plow', 'plunger', 'Polaroid camera', 'pole', 'police van',
'poncho', 'pool table', 'pop bottle', 'pot', 'potter's wheel', 'power drill',
'prayer rug', 'printer', 'prison', 'projectile', 'projector', 'puck', 'punching
bag', 'purse', 'quill', 'quilt', 'racer', 'racket', 'radiator', 'radio', 'radio
telescope', 'rain barrel', 'recreational vehicle', 'reel', 'reflex camera',
'refrigerator', 'remote control', 'restaurant', 'revolver', 'rifle', 'rocking
chair', 'rotisserie', 'rubber eraser', 'rugby ball', 'rule', 'running shoe',
'safe', 'safety pin', 'saltshaker', 'sandal', 'sarong', 'sax', 'scabbard',
'scale', 'school bus', 'schooner', 'scoreboard', 'screen', 'screw',
'screwdriver', 'seat belt', 'sewing machine', 'shield', 'shoe shop', 'shoji',
'shopping basket', 'shopping cart', 'shovel', 'shower cap', 'shower curtain',
'ski', 'ski mask', 'sleeping bag', 'slide rule', 'sliding door', 'slot',
'snorkel', 'snowmobile', 'snowplow', 'soap dispenser', 'soccer ball', 'sock',
'solar dish', 'sombbrero', 'soup bowl', 'space bar', 'space heater', 'space
shuttle', 'spatula', 'speedboat', 'spider web', 'spindle', 'sports car',
'spotlight', 'stage', 'steam locomotive', 'steel arch bridge', 'steel drum',
'stethoscope', 'stole', 'stone wall', 'stopwatch', 'stove', 'strainer',
'streetcar', 'stretcher', 'studio couch', 'stupa', 'submarine', 'suit',
'sundial', 'sunglass', 'sunglasses', 'sunscreen', 'suspension bridge', 'swab',
'sweatshirt', 'swimming trunks', 'swing', 'switch', 'syringe', 'table lamp',
'tank', 'tape player', 'teapot', 'teddy', 'television', 'tennis ball', 'thatch',
'theater curtain', 'thimble', 'thresher', 'throne', 'tile roof', 'toaster',

'tobacco shop', 'toilet seat', 'torch', 'totem pole', 'tow truck', 'toyshop', 'tractor', 'trailer truck', 'tray', 'trench coat', 'tricycle', 'trimaran', 'tripod', 'triumphal arch', 'trolleybus', 'trombone', 'tub', 'turnstile', 'typewriter keyboard', 'umbrella', 'unicycle', 'upright', 'vacuum', 'vase', 'vault', 'velvet', 'vending machine', 'vestment', 'viaduct', 'violin', 'volleyball', 'waffle iron', 'wall clock', 'wallet', 'wardrobe', 'warplane', 'washbasin', 'washer', 'water bottle', 'water jug', 'water tower', 'whiskey jug', 'whistle', 'wig', 'window screen', 'window shade', 'Windsor tie', 'wine bottle', 'wing', 'wok', 'wooden spoon', 'wool', 'worm fence', 'wreck', 'yaw', 'yurt', 'web site', 'comic book', 'crossword puzzle', 'street sign', 'traffic light', 'book jacket', 'menu', 'plate', 'guacamole', 'consomme', 'hot pot', 'trifle', 'ice cream', 'ice lolly', 'French loaf', 'bagel', 'pretzel', 'cheeseburger', 'hotdog', 'mashed potato', 'head cabbage', 'broccoli', 'cauliflower', 'zucchini', 'spaghetti squash', 'acorn squash', 'butternut squash', 'cucumber', 'artichoke', 'bell pepper', 'cardoon', 'mushroom', 'Granny Smith', 'strawberry', 'orange', 'lemon', 'fig', 'pineapple', 'banana', 'jackfruit', 'custard apple', 'pomegranate', 'hay', 'carbonara', 'chocolate sauce', 'dough', 'meat loaf', 'pizza', 'potpie', 'burrito', 'red wine', 'espresso', 'cup', 'eggnog', 'alp', 'bubble', 'cliff', 'coral reef', 'geyser', 'lakeside', 'promontory', 'sandbar', 'seashore', 'valley', 'volcano', 'ballplayer', 'groom', 'scuba diver', 'rapeseed', 'daisy', 'yellow lady's slipper', 'corn', 'acorn', 'hip', 'buckeye', 'coral fungus', 'agaric', 'gyromitra', 'stinkhorn', 'earthstar', 'hen-of-the-woods', 'bolete', 'ear', 'toilet tissue']

- Predict on the loaded image

```
[7]: # Make prediction
with torch.no_grad():
    predictions = model(torch.unsqueeze(xi, 0))

probability = torch.softmax(predictions, dim=-1)
values, indices = torch.topk(probability, k=3, dim=1)
top_classes = [class_names[idx] for idx in indices.squeeze()]

for v, c in zip(values.numpy().tolist()[0], top_classes):
    print((c, v))
```

```
('Indian elephant', 0.8288949131965637)
('tusker', 0.1690331995487213)
('African elephant', 0.0020219553261995316)
```

39 Pre-trained Image Features

- The network is trained on ImageNet, which contains many classes and images.
- The learned features generalize well to other tasks
 - capture high-level discriminative information about objects



40 Computing Image Features

- Use the pre-trained network as a feature extractor
 - remove the last layer (the classifier)
 - apply average pooling on the feature map
 - * $7 \times 7 \times 2048 \rightarrow 1 \times 2048$

```
[11]: # create an instance of the model w/o the last layer
model_f = torch.nn.Sequential(*list(model.children())[:-1]) # Remove last layer
model_f = torch.nn.Sequential(
    model_f,
    torch.nn.AdaptiveAvgPool2d((1, 1)), # Global average pooling
    torch.nn.Flatten() # Flatten to [batch_size, features]
)
model_f.eval()

print("Model structure:")
print(model_f)

# Compute features (equivalent to model_f.predict(Xim))
with torch.no_grad():
    Xf = model_f(Xim) # Xim should be a PyTorch tensor [batch_size, 3, H, W]

print(f"Feature shape: {Xf.shape}") # Should be [batch_size, 2048]
```

```
Model structure:
Sequential(
  (0): Sequential(
    (0): Conv2d(3, 64, kernel_size=(7, 7), stride=(2, 2), padding=(3, 3),
    bias=False)
```

```

        (1): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (2): ReLU(inplace=True)
        (3): MaxPool2d(kernel_size=3, stride=2, padding=1, dilation=1,
ceil_mode=False)
        (4): Sequential(
          (0): Bottleneck(
            (conv1): Conv2d(64, 64, kernel_size=(1, 1), stride=(1, 1), bias=False)
            (bn1): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
            (conv2): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
            (bn2): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
            (conv3): Conv2d(64, 256, kernel_size=(1, 1), stride=(1, 1), bias=False)
            (bn3): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
            (relu): ReLU(inplace=True)
            (downsample): Sequential(
              (0): Conv2d(64, 256, kernel_size=(1, 1), stride=(1, 1), bias=False)
              (1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
            )
          )
        (1): Bottleneck(
          (conv1): Conv2d(256, 64, kernel_size=(1, 1), stride=(1, 1), bias=False)
          (bn1): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
          (conv2): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn2): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
          (conv3): Conv2d(64, 256, kernel_size=(1, 1), stride=(1, 1), bias=False)
          (bn3): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
          (relu): ReLU(inplace=True)
        )
        (2): Bottleneck(
          (conv1): Conv2d(256, 64, kernel_size=(1, 1), stride=(1, 1), bias=False)
          (bn1): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
          (conv2): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn2): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
          (conv3): Conv2d(64, 256, kernel_size=(1, 1), stride=(1, 1), bias=False)
          (bn3): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)

```



```

        (relu): ReLU(inplace=True)
    )
)
(5): Sequential(
  (0): Bottleneck(
    (conv1): Conv2d(256, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
    (bn1): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (conv2): Conv2d(128, 128, kernel_size=(3, 3), stride=(2, 2), padding=(1,
1), bias=False)
    (bn2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (conv3): Conv2d(128, 512, kernel_size=(1, 1), stride=(1, 1), bias=False)
    (bn3): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (relu): ReLU(inplace=True)
    (downsample): Sequential(
      (0): Conv2d(256, 512, kernel_size=(1, 1), stride=(2, 2), bias=False)
      (1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    )
  )
  (1): Bottleneck(
    (conv1): Conv2d(512, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
    (bn1): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (conv2): Conv2d(128, 128, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
    (bn2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (conv3): Conv2d(128, 512, kernel_size=(1, 1), stride=(1, 1), bias=False)
    (bn3): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (relu): ReLU(inplace=True)
  )
  (2): Bottleneck(
    (conv1): Conv2d(512, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
    (bn1): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (conv2): Conv2d(128, 128, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
    (bn2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (conv3): Conv2d(128, 512, kernel_size=(1, 1), stride=(1, 1), bias=False)
    (bn3): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (relu): ReLU(inplace=True)
  )
)

```

```

        (3): Bottleneck(
          (conv1): Conv2d(512, 128, kernel_size=(1, 1), stride=(1, 1), bias=False)
          (bn1): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
          (conv2): Conv2d(128, 128, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
          (bn2): BatchNorm2d(128, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
          (conv3): Conv2d(128, 512, kernel_size=(1, 1), stride=(1, 1), bias=False)
          (bn3): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
          (relu): ReLU(inplace=True)
        )
      )
    (6): Sequential(
      (0): Bottleneck(
        (conv1): Conv2d(512, 256, kernel_size=(1, 1), stride=(1, 1), bias=False)
        (bn1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (conv2): Conv2d(256, 256, kernel_size=(3, 3), stride=(2, 2), padding=(1,
1), bias=False)
        (bn2): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (conv3): Conv2d(256, 1024, kernel_size=(1, 1), stride=(1, 1),
bias=False)
        (bn3): BatchNorm2d(1024, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (relu): ReLU(inplace=True)
        (downsample): Sequential(
          (0): Conv2d(512, 1024, kernel_size=(1, 1), stride=(2, 2), bias=False)
          (1): BatchNorm2d(1024, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        )
      )
      (1): Bottleneck(
        (conv1): Conv2d(1024, 256, kernel_size=(1, 1), stride=(1, 1),
bias=False)
        (bn1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (conv2): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
        (bn2): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (conv3): Conv2d(256, 1024, kernel_size=(1, 1), stride=(1, 1),
bias=False)
        (bn3): BatchNorm2d(1024, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
        (relu): ReLU(inplace=True)
      )
    )
  )
)

```

```

    )
    (2): Bottleneck(
      (conv1): Conv2d(1024, 256, kernel_size=(1, 1), stride=(1, 1),
bias=False)
      (bn1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
      (conv2): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
      (bn2): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
      (conv3): Conv2d(256, 1024, kernel_size=(1, 1), stride=(1, 1),
bias=False)
      (bn3): BatchNorm2d(1024, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
      (relu): ReLU(inplace=True)
    )
    (3): Bottleneck(
      (conv1): Conv2d(1024, 256, kernel_size=(1, 1), stride=(1, 1),
bias=False)
      (bn1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
      (conv2): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
      (bn2): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
      (conv3): Conv2d(256, 1024, kernel_size=(1, 1), stride=(1, 1),
bias=False)
      (bn3): BatchNorm2d(1024, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
      (relu): ReLU(inplace=True)
    )
    (4): Bottleneck(
      (conv1): Conv2d(1024, 256, kernel_size=(1, 1), stride=(1, 1),
bias=False)
      (bn1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
      (conv2): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
      (bn2): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
      (conv3): Conv2d(256, 1024, kernel_size=(1, 1), stride=(1, 1),
bias=False)
      (bn3): BatchNorm2d(1024, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
      (relu): ReLU(inplace=True)
    )
    (5): Bottleneck(
      (conv1): Conv2d(1024, 256, kernel_size=(1, 1), stride=(1, 1),

```

```

bias=False)
    (bn1): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (conv2): Conv2d(256, 256, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
    (bn2): BatchNorm2d(256, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (conv3): Conv2d(256, 1024, kernel_size=(1, 1), stride=(1, 1),
bias=False)
    (bn3): BatchNorm2d(1024, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (relu): ReLU(inplace=True)
    )
    )
    (7): Sequential(
    (0): Bottleneck(
    (conv1): Conv2d(1024, 512, kernel_size=(1, 1), stride=(1, 1),
bias=False)
    (bn1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (conv2): Conv2d(512, 512, kernel_size=(3, 3), stride=(2, 2), padding=(1,
1), bias=False)
    (bn2): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (conv3): Conv2d(512, 2048, kernel_size=(1, 1), stride=(1, 1),
bias=False)
    (bn3): BatchNorm2d(2048, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (relu): ReLU(inplace=True)
    (downsample): Sequential(
    (0): Conv2d(1024, 2048, kernel_size=(1, 1), stride=(2, 2), bias=False)
    (1): BatchNorm2d(2048, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    )
    )
    (1): Bottleneck(
    (conv1): Conv2d(2048, 512, kernel_size=(1, 1), stride=(1, 1),
bias=False)
    (bn1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (conv2): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
    (bn2): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (conv3): Conv2d(512, 2048, kernel_size=(1, 1), stride=(1, 1),
bias=False)
    (bn3): BatchNorm2d(2048, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)

```

```

        (relu): ReLU(inplace=True)
    )
    (2): Bottleneck(
      (conv1): Conv2d(2048, 512, kernel_size=(1, 1), stride=(1, 1),
bias=False)
      (bn1): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
      (conv2): Conv2d(512, 512, kernel_size=(3, 3), stride=(1, 1), padding=(1,
1), bias=False)
      (bn2): BatchNorm2d(512, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
      (conv3): Conv2d(512, 2048, kernel_size=(1, 1), stride=(1, 1),
bias=False)
      (bn3): BatchNorm2d(2048, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
      (relu): ReLU(inplace=True)
    )
  )
  (8): AdaptiveAvgPool2d(output_size=(1, 1))
)
(1): AdaptiveAvgPool2d(output_size=(1, 1))
(2): Flatten(start_dim=1, end_dim=-1)
)
Feature shape: torch.Size([200, 2048])

```

- Now use the features to train a binary classifier
- First setup the dataset

```

[12]: # randomly split data into train and test set
( trainXf, testXf,      # features
  trainY, testY,      # class labels
  trainXim, testXim,   # pre-processed images
  trainXraw, testXraw  # original images
) = \
  model_selection.train_test_split(Xf, Y, Xim, Xraw,
    train_size=0.8, test_size=0.2, random_state=4487)

trainX = trainXim.reshape((trainXim.shape[0], -1))
testX = testXim.reshape((testXim.shape[0], -1))

print(trainXf.shape)
print(testXf.shape)
print(trainXim.shape)
print(trainX.shape)

```

```

torch.Size([160, 2048])
torch.Size([40, 2048])
torch.Size([160, 3, 224, 224])

```

```
torch.Size([160, 150528])
```

- Use the image features to train a standard SVM classifier

```
[13]: # setup the list of parameters to try
paramgrid = {'C': logspace(-3,3,13)}
print(paramgrid)

# setup cross-validation
svmcv = model_selection.GridSearchCV(svm.SVC(kernel='linear'), paramgrid, cv=5,
                                     n_jobs=-1, verbose=True)

# run cross-validation (train for each split)
svmcv.fit(trainXf.numpy(), trainY);
```

```
{'C': array([1.00000000e-03, 3.16227766e-03, 1.00000000e-02, 3.16227766e-02,
            1.00000000e-01, 3.16227766e-01, 1.00000000e+00, 3.16227766e+00,
            1.00000000e+01, 3.16227766e+01, 1.00000000e+02, 3.16227766e+02,
            1.00000000e+03])}
```

Fitting 5 folds for each of 13 candidates, totalling 65 fits

- Predict with the classifier
- The accuracy is great!

```
[14]: # predict
predY = svmcv.predict(testXf.numpy())

acc = metrics.accuracy_score(testY, predY)
print("test accuracy = " + str(acc))
```

test accuracy = 1.0

- For comparison, also train an SVM on the raw images
- The accuracy is really bad (almost chance level)

```
[15]: svmcv_im = model_selection.GridSearchCV(svm.SVC(kernel='linear'), paramgrid,
      ↪cv=5,
                                     n_jobs=-1, verbose=True)

# run cross-validation (train for each split)
svmcv_im.fit(trainX.numpy(), trainY);

# Directly use svmcv to make predictions
predY = svmcv_im.predict(testX)

acc = metrics.accuracy_score(testY, predY)
print("test accuracy = " + str(acc))
```

Fitting 5 folds for each of 13 candidates, totalling 65 fits

test accuracy = 0.55

41 Outline

- Going deeper
 - ReLU and Batchnorm
- Optimization methods
- Deep architectures and Image classification
- **Transfer learning**

42 Transfer learning and fine-tuning

- The pre-trained network can also be “fine-tuned” for a new image classification task.
 - This is called transfer learning.
- We could:
 - 1) start from the pre-trained network, and train the whole network on the new classification task
 - 2) Rather than retrain the whole network
 - fix the lower layers that extract features (no need to train them since they are good features)
 - train the last few layers to perform the new task
- Does not need as much data, compared to the original ImageNet.
- First, let’s see ResNet50 predicts on our images

```
[16]: # try the original ResNet50
testYscore = model(testXim)

# get the class labels
predY = torch.argmax(testYscore, dim=1).tolist()
predY = [class_names[i] for i in predY]
testYcl = le.inverse_transform(testY)
```

```
[17]: # make titles
titles = []
for i in range(len(predY)):
    titles.append(predY[i] + "\n(" + testYcl[i] + ")")
```

```
[18]: # make a plot: predicted (true)
plt.figure(figsize=(9,20))
show_imgs(testXraw, nc=5, titles=titles)
```

ice cream
(ice_cream)



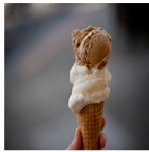
ice cream
(frozen_yogurt)



wooden spoon
(frozen_yogurt)



ice cream
(ice_cream)



chocolate sauce
(frozen_yogurt)



mixing bowl
(frozen_yogurt)



consomme
(frozen_yogurt)



ice cream
(ice_cream)



strawberry
(frozen_yogurt)



plate
(ice_cream)



plate
(frozen_yogurt)



cup
(frozen_yogurt)



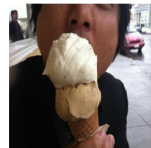
ice cream
(frozen_yogurt)



butternut squash
(frozen_yogurt)



ice cream
(ice_cream)



ice cream
(ice_cream)



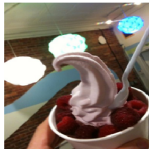
ice cream
(ice_cream)



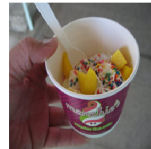
pot
(frozen_yogurt)



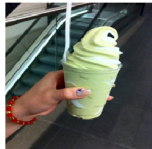
ice cream
(frozen_yogurt)



consomme
(frozen_yogurt)



ice cream
(frozen_yogurt)



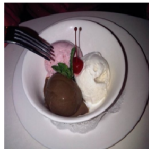
ice cream
(ice_cream)



trifle
(ice_cream)



ice cream
(ice_cream)



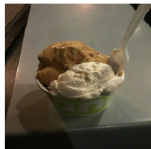
ice cream
(frozen_yogurt)



chocolate sauce
(ice_cream)



ice cream
(ice_cream)



consomme
(frozen_yogurt)



ice cream
(ice_cream)



ice cream
(ice_cream)



ice cream
(ice_cream)



ice cream
(ice_cream)



consomme
(frozen_yogurt)



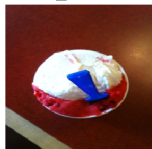
burrito
(ice_cream)



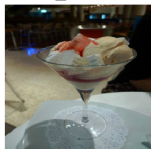
ice cream
(frozen_yogurt)



bagel
(ice_cream)



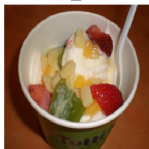
eggnog
(ice_cream)



consomme
(frozen_yogurt)



ice cream
(frozen_yogurt)



pick
(ice_cream)



- We will use the `Model` class
 - allows for more complex layer interactions than `Sequential`
 - first, connect layers one-by-one
 - * instantiate the layer (with hyperparameters)
 - * then pass the previous layer to it.
- Second, create the `Model`
 - specify the input and output layers
 - the network is everything in between
- Fix the weights of the layers of `ResNet` so they will not change during training

```
[19]: # Add global average pooling and custom classifier
def build_toponly(num_classes):
    # Load pre-trained ResNet50
    model = models.resnet50(weights=models.ResNet50_Weights.IMAGENET1K_V1)

    # Freeze all layers (equivalent to layer.trainable = False)
    for param in model.parameters():
        param.requires_grad = False

    model_f = torch.nn.Sequential(*list(model.children())[:-1]) # Remove last
    ↪ layer
    base_model = torch.nn.Sequential(
        model_f,
        torch.nn.AdaptiveAvgPool2d((1, 1)), # Global average pooling
        torch.nn.Flatten(), # Flatten to [batch_size, features]
        nn.Linear(2048, 32), # Fully-connected layer (1x32)
        nn.ReLU(inplace=True), # ReLU activation
        nn.Linear(32, num_classes), # Final classifier (2 classes)
    )

    return base_model
```

- Setup validation set for monitoring

```
[20]: # generate a fixed validation set using 10% of the training set
_, validXim, vtrainXraw, _, vtrainY, validY = \
    model_selection.train_test_split(trainXim, trainXraw, trainY,
    train_size=0.9, test_size=0.1, random_state=6487)
print('Training set: {}'.format(len(trainXraw), trainXraw[0].size)
print(validXim.size())
```

```
Training set: 160 (224, 224)
torch.Size([16, 3, 224, 224])
```

- Setup the data augmentation generator

```
[21]: class GaussianNoise:
    """Add Gaussian noise with zero mean and specified variance"""
    def __init__(self, sigma2=0.1):
        self.sigma2 = sigma2

    def __call__(self, x):
        return x + torch.randn_like(x) * self.sigma2

train_transform = transforms.Compose([
    transforms.RandomResizedCrop(224),
    transforms.RandomRotation(10),
    transforms.RandomHorizontalFlip(0.5),
    # transforms.RandomAffine(
    #     degrees=0,
    #     translate=(0.1, 0.1), # width_shift_range and height_shift_range
    #     scale=(0.9, 1.1),    # zoom_range
    #     shear=0.1           # shear_range
    # ),
    transforms.ToTensor(),
    # GaussianNoise(),
    transforms.Normalize(mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]),
])

class MyDataset(torch.utils.data.Dataset):
    def __init__(self, data, targets, transform):
        self.data = data
        self.targets = targets
        self.transform = transform

    def __len__(self):
        return len(self.data)

    def __getitem__(self, idx):
        x = self.data[idx]
        x = self.transform(x)
        y = self.targets[idx]

        return x, y

# Create datasets with augmentation
trainset = MyDataset(vtrainXraw, torch.LongTensor(vtrainY),
    ↪transform=train_transform)
validset = TensorDataset(validXim, torch.LongTensor(validY))
```

- Train the model
 - only the last layers that we added are updated

```
[22]: # Early stopping implementation
class EarlyStopping:
    def __init__(self, patience=5, min_delta=0.0001, verbose=1):
        self.patience = patience
        self.min_delta = min_delta
        self.counter = 0
        self.best_accuracy = 0.0
        self.early_stop = False
        self.verbose = verbose

    def __call__(self, current_accuracy):
        if current_accuracy - self.best_accuracy > self.min_delta:
            self.best_accuracy = current_accuracy
            self.counter = 0
        else:
            self.counter += 1
            if self.verbose:
                print(f'EarlyStopping counter: {self.counter} out of {self.
→patience}')
            if self.counter >= self.patience:
                self.early_stop = True
        return self.early_stop
```

```
[23]: # Model Parameter Summarization
def summarize_layer_parameters(model):
    """
    Summarizes the number of parameters in each layer of a PyTorch model.
    """
    trainable_table = PrettyTable(["Module Name", "Parameters"])
    untrainable_table = PrettyTable(["Module Name", "Parameters"])
    total_params = {'train':0, 'untrain':0}
    for name, parameter in model.named_parameters():
        if parameter.requires_grad: # Only count trainable parameters
            params = parameter.numel()
            trainable_table.add_row([name, params])
            total_params['train'] += params
        else: # Only count untrainable parameters
            params = parameter.numel()
            untrainable_table.add_row([name, params])
            total_params['untrain'] += params

    print(trainable_table)
    print(f"\nTotal Trainable Parameters: {total_params['train']}")
    if total_params['untrain'] > 0:
        print(untrainable_table)
    print(f"\nTotal Un-Trainable Parameters: {total_params['untrain']}")
```

```

[24]: # Model Evaluation
def evaluate_model(model, eval_loader, criterion=None):
    model.eval()
    eval_correct = 0
    eval_total = 0
    eval_loss = 0.0

    with torch.no_grad():
        for data, target in eval_loader:
            data, target = data.to(device), target.to(device)
            output = model(data)
            _, predicted = torch.max(output.data, 1)
            eval_total += target.size(0)
            eval_correct += (predicted == target).sum().item()

            if criterion is not None:
                loss = criterion(output, target)
                eval_loss += loss.item()

    eval_accuracy = 100 * eval_correct / eval_total
    avg_eval_loss = eval_loss / len(eval_loader)

    return eval_accuracy if criterion is None else (eval_accuracy,
    ↪avg_eval_loss)

# Model Training
def train_model(model, train_loader, valid_loader, criterion, optimizer,
    ↪epochs=100, patience=5, batch_scheduler=None, epoch_scheduler=None):
    history = {
        'train_loss': [], 'val_loss': [],
        'train_accuracy': [], 'val_accuracy': [],
        'learning_rate': []
    }
    early_stopping = EarlyStopping(patience=patience)
    assert batch_scheduler is None or epoch_scheduler is None

    for epoch in range(epochs):
        # Training phase
        model.train()
        train_loss = 0.0
        train_correct = 0
        train_total = 0

        for batch_idx, (data, target) in enumerate(train_loader):
            data, target = data.to(device), target.to(device)

```

```

optimizer.zero_grad()
output = model(data)
loss = criterion(output, target)

loss.backward()
optimizer.step()
if batch_scheduler is not None:
    batch_scheduler.step()
    history['learning_rate'].append(optimizer.
↪state_dict()['param_groups'][0]['lr'])

train_loss += loss.item()
_, predicted = torch.max(output.data, 1)
train_total += target.size(0)
train_correct += (predicted == target).sum().item()

train_accuracy = 100 * train_correct / train_total
avg_train_loss = train_loss / len(train_loader)

# Validation phase
val_accuracy, avg_val_loss = ↪
↪evaluate_model(model, valid_loader, criterion)

# Store history
history['train_loss'].append(avg_train_loss)
history['val_loss'].append(avg_val_loss)
history['train_accuracy'].append(train_accuracy / 100)
history['val_accuracy'].append(val_accuracy / 100)

train_msg = f' Train Loss: {avg_train_loss:.6f}, Train Acc: ↪
↪{train_accuracy:.2f}%;'
valid_msg = f' Val Loss: {avg_val_loss:.6f}, Val Acc: {val_accuracy:.
↪2f}%'
```

```

print(f'Epoch {epoch+1}/{epochs}:', train_msg, valid_msg)

# Check early stopping
if patience > 0:
    early_stopping(val_accuracy)
    if early_stopping.early_stop:
        print("Early stopping triggered")
        break
    else:
        torch.save(model.state_dict(), 'best_model.pth')
else:
    torch.save(model.state_dict(), 'best_model.pth')
if epoch_scheduler is not None:

```

```

        if isinstance(epoch_scheduler, ReduceLROnPlateau):
            epoch_scheduler.step(avg_val_loss)
        else:
            epoch_scheduler.step()
            history['learning_rate'].append(optimizer.
↪state_dict()['param_groups'][0]['lr'])

    # Load best model
    model.load_state_dict(torch.load('best_model.pth'))
    return history, model

```

```

[25]: # Initialize random seed
torch.manual_seed(4487)
random.seed(4487)
np.random.seed(4487)
if torch.cuda.is_available():
    torch.cuda.manual_seed(4487)
    torch.backends.cudnn.deterministic = True

# Create model instance
ftmodel = build_toponly(2)

# Print model architecture
# summarize_layer_parameters(ftmodel)

# Move model to device (GPU if available)
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
ftmodel = ftmodel.to(device)
print(f"Using device: {device}")

# Create DataLoaders
batch_size = 50

train_loader = DataLoader(trainset, batch_size=batch_size, shuffle=True)
valid_loader = DataLoader(validset, batch_size=16, shuffle=False)

# Compile the network (equivalent in PyTorch)
criterion = nn.CrossEntropyLoss()
train_param = filter(lambda p: p.requires_grad, ftmodel.parameters())
optimizer = optim.RMSprop(ftmodel.parameters(), lr=0.001)

# Train the model
history, ftmodel = train_model(ftmodel, train_loader, valid_loader, criterion,
↪optimizer, epochs=50, patience=-1)

```

Using device: cuda

Epoch 1/50: Train Loss: 2.448295, Train Acc: 50.00%; Val Loss: 1.203903, Val

Acc: 37.50%
 Epoch 2/50: Train Loss: 0.730163, Train Acc: 64.58%; Val Loss: 0.580642, Val
 Acc: 62.50%
 Epoch 3/50: Train Loss: 0.673592, Train Acc: 55.56%; Val Loss: 0.571283, Val
 Acc: 81.25%
 Epoch 4/50: Train Loss: 0.541362, Train Acc: 85.42%; Val Loss: 0.544886, Val
 Acc: 81.25%
 Epoch 5/50: Train Loss: 0.550259, Train Acc: 68.75%; Val Loss: 0.504507, Val
 Acc: 75.00%
 Epoch 6/50: Train Loss: 0.471064, Train Acc: 86.81%; Val Loss: 0.544714, Val
 Acc: 87.50%
 Epoch 7/50: Train Loss: 0.469768, Train Acc: 81.94%; Val Loss: 0.484918, Val
 Acc: 81.25%
 Epoch 8/50: Train Loss: 0.442836, Train Acc: 85.42%; Val Loss: 0.629478, Val
 Acc: 81.25%
 Epoch 9/50: Train Loss: 0.454876, Train Acc: 79.86%; Val Loss: 0.442438, Val
 Acc: 75.00%
 Epoch 10/50: Train Loss: 0.508065, Train Acc: 75.00%; Val Loss: 0.434009,
 Val Acc: 68.75%
 Epoch 11/50: Train Loss: 0.371479, Train Acc: 90.28%; Val Loss: 0.423633,
 Val Acc: 75.00%
 Epoch 12/50: Train Loss: 0.388276, Train Acc: 87.50%; Val Loss: 0.423423,
 Val Acc: 75.00%
 Epoch 13/50: Train Loss: 0.372913, Train Acc: 87.50%; Val Loss: 0.437093,
 Val Acc: 87.50%
 Epoch 14/50: Train Loss: 0.333709, Train Acc: 88.19%; Val Loss: 0.412725,
 Val Acc: 75.00%
 Epoch 15/50: Train Loss: 0.422438, Train Acc: 82.64%; Val Loss: 0.486512,
 Val Acc: 68.75%
 Epoch 16/50: Train Loss: 0.383001, Train Acc: 84.72%; Val Loss: 0.494848,
 Val Acc: 68.75%
 Epoch 17/50: Train Loss: 0.349522, Train Acc: 88.89%; Val Loss: 0.394826,
 Val Acc: 81.25%
 Epoch 18/50: Train Loss: 0.289626, Train Acc: 90.97%; Val Loss: 0.386807,
 Val Acc: 75.00%
 Epoch 19/50: Train Loss: 0.340366, Train Acc: 85.42%; Val Loss: 0.440302,
 Val Acc: 87.50%
 Epoch 20/50: Train Loss: 0.444004, Train Acc: 76.39%; Val Loss: 0.388452,
 Val Acc: 75.00%
 Epoch 21/50: Train Loss: 0.316627, Train Acc: 89.58%; Val Loss: 0.465533,
 Val Acc: 87.50%
 Epoch 22/50: Train Loss: 0.327823, Train Acc: 90.97%; Val Loss: 0.525889,
 Val Acc: 68.75%
 Epoch 23/50: Train Loss: 0.414128, Train Acc: 75.69%; Val Loss: 0.408272,
 Val Acc: 81.25%
 Epoch 24/50: Train Loss: 0.265495, Train Acc: 90.97%; Val Loss: 0.387755,
 Val Acc: 75.00%
 Epoch 25/50: Train Loss: 0.256789, Train Acc: 93.06%; Val Loss: 0.388530,

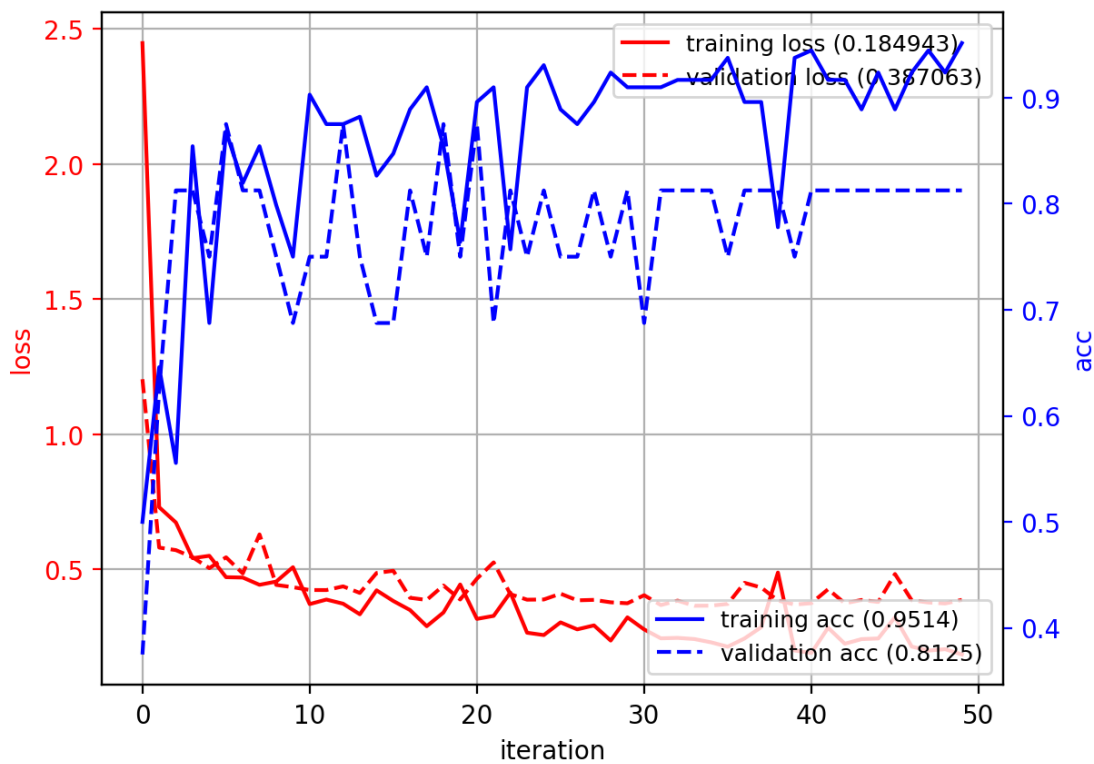
Val Acc: 81.25%
 Epoch 26/50: Train Loss: 0.303804, Train Acc: 88.89%; Val Loss: 0.410055,
 Val Acc: 75.00%
 Epoch 27/50: Train Loss: 0.277904, Train Acc: 87.50%; Val Loss: 0.385395,
 Val Acc: 75.00%
 Epoch 28/50: Train Loss: 0.292617, Train Acc: 89.58%; Val Loss: 0.387150,
 Val Acc: 81.25%
 Epoch 29/50: Train Loss: 0.236481, Train Acc: 92.36%; Val Loss: 0.378102,
 Val Acc: 75.00%
 Epoch 30/50: Train Loss: 0.322384, Train Acc: 90.97%; Val Loss: 0.374462,
 Val Acc: 81.25%
 Epoch 31/50: Train Loss: 0.277709, Train Acc: 90.97%; Val Loss: 0.404338,
 Val Acc: 68.75%
 Epoch 32/50: Train Loss: 0.244749, Train Acc: 90.97%; Val Loss: 0.367918,
 Val Acc: 81.25%
 Epoch 33/50: Train Loss: 0.246381, Train Acc: 91.67%; Val Loss: 0.384215,
 Val Acc: 81.25%
 Epoch 34/50: Train Loss: 0.242065, Train Acc: 91.67%; Val Loss: 0.365272,
 Val Acc: 81.25%
 Epoch 35/50: Train Loss: 0.229985, Train Acc: 91.67%; Val Loss: 0.365664,
 Val Acc: 81.25%
 Epoch 36/50: Train Loss: 0.213786, Train Acc: 93.75%; Val Loss: 0.372112,
 Val Acc: 75.00%
 Epoch 37/50: Train Loss: 0.244127, Train Acc: 89.58%; Val Loss: 0.450276,
 Val Acc: 81.25%
 Epoch 38/50: Train Loss: 0.285872, Train Acc: 89.58%; Val Loss: 0.432847,
 Val Acc: 81.25%
 Epoch 39/50: Train Loss: 0.488335, Train Acc: 77.78%; Val Loss: 0.384730,
 Val Acc: 81.25%
 Epoch 40/50: Train Loss: 0.199557, Train Acc: 93.75%; Val Loss: 0.369781,
 Val Acc: 75.00%
 Epoch 41/50: Train Loss: 0.190675, Train Acc: 94.44%; Val Loss: 0.374773,
 Val Acc: 81.25%
 Epoch 42/50: Train Loss: 0.283932, Train Acc: 91.67%; Val Loss: 0.426474,
 Val Acc: 81.25%
 Epoch 43/50: Train Loss: 0.224703, Train Acc: 91.67%; Val Loss: 0.374384,
 Val Acc: 81.25%
 Epoch 44/50: Train Loss: 0.241796, Train Acc: 88.89%; Val Loss: 0.386701,
 Val Acc: 81.25%
 Epoch 45/50: Train Loss: 0.244265, Train Acc: 92.36%; Val Loss: 0.379429,
 Val Acc: 81.25%
 Epoch 46/50: Train Loss: 0.322765, Train Acc: 88.89%; Val Loss: 0.482789,
 Val Acc: 81.25%
 Epoch 47/50: Train Loss: 0.214185, Train Acc: 92.36%; Val Loss: 0.385663,
 Val Acc: 81.25%
 Epoch 48/50: Train Loss: 0.199305, Train Acc: 94.44%; Val Loss: 0.377430,
 Val Acc: 81.25%
 Epoch 49/50: Train Loss: 0.204517, Train Acc: 92.36%; Val Loss: 0.373763,

Val Acc: 81.25%

Epoch 50/50: Train Loss: 0.184943, Train Acc: 95.14%; Val Loss: 0.387063, Val Acc: 81.25%

- Training/validation curves

```
[26]: # Plot the training history
plot_history(history)
```



- Compute test accuracy

```
[27]: ftmodel.eval()
predYscore = ftmodel(testXim.to(device))
predY = torch.argmax(predYscore,dim=-1).cpu().numpy()
acc = metrics.accuracy_score(testY, predY)
print("test accuracy:", acc)
```

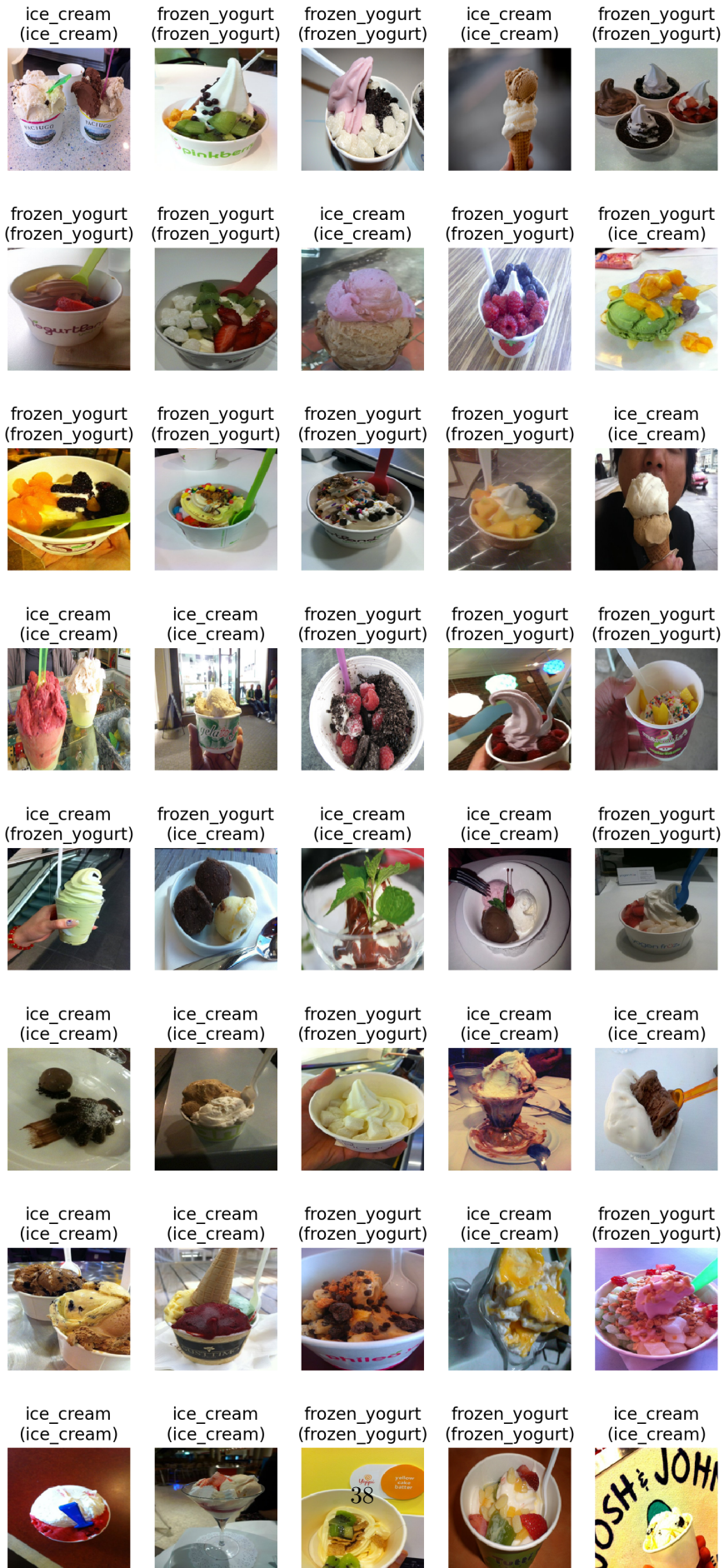
test accuracy: 0.925

- Visualize the predictions

```
[28]: # get the class labels
predYcl = le.inverse_transform(predY)
# make titles
```

```
titles = []  
for i in range(len(predYcl)):  
    titles.append(predYcl[i] + "\n(" + testYcl[i] + ")")
```

```
[29]: # make a plot: predicted (true)  
plt.figure(figsize=(9,20))  
show_imgs(testXraw, nc=5, titles=titles)
```



43 Generalization: Using more features

- the last layers of ResNet are high-level semantic features, which could be complemented by low-level or mid-level features.
- extract features from multiple layers and concatenate them.
- However, you may still get similar accuracy when using more features

44 Other tips

- If the dataset is too large to load into memory, can use `tf.keras.utils.image_dataset_from_directory` to build a dataset from directory.
 - will handle disk reading and caching of files.
- each pre-trained network has its own `preprocess_input` function.
 - this function can also be used as a layer if you want to embed preprocessing into the network.
 - the expected input (pixel range and image size) varies - check the documentation.

45 Summary

- Network “tricks” to go deeper:
 - ReLU activations - encourage sparse activations, reduce vanishing gradient problem.
 - Batch Normalization - reparameterizing the network to be more stable.
- Optimization “tricks”:
 - SGD with decaying learning rate
 - * analysis shows that noisy gradient actually helps to ignore the small local minima.
 - Adaptive learning rates: AdaGrad, RMSProp, Adam.
- Deep architectures:
 - advances of deep learning has been driven by the ImageNet competition.
 - error rate decreases as the depth increases.
 - as depth increases, need to have a smart architecture design to make training more effective.
 - * VGGnet - deep version of standard CNN
 - * InceptionNet - auxiliary tasks
 - * ResNet - residual learning, equivalent to an ensemble of models.
 - * SEnet - channel-wise attention
 - Vision Transformer & self-attention
 - * more training data can trump the inductive bias in CNNs
 - * (at the cost of computation and parameters)
- Transfer learning
 - feature extractors learned from ImageNet can be re-used for other tasks.
 - * global pooled features for high-level tasks
 - * mid-level and high-level feature maps for other tasks like localization and detection.

- fine-tune a pre-trained network on our own specific task.
 - * use the pre-trained network as a “backbone” network.
- Pre-trained models: <https://keras.io/api/applications/>